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Review

Advancements in Substrate Technologies and Design Challenges for 5G mmWave Systems

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Abstract

Substrates have become essential enabling materials for creating lightweight electronic components, particularly supporting advanced telecommunication technologies. This progress is driven by continuous advancements in novel substrate materials and cutting-edge fabrication techniques, pushing the limits of high-frequency device design. This paper explores both the challenges and breakthroughs in 5G mmWave substrate technology, focusing on recent developments in materials, device fabrication, and integration methods that enhance performance and provide an in-depth analysis of the importance of mmWave technology. This paper also highlights the key concerns in substrate design for researchers and academicians to accelerate the invention and commercialization of substrate designs in areas such as antenna engineering and integrated circuit technologies, as well as addressing key issues like scalability, thermal impact in flexible substrates, and AI-driven RSSI-aware beamforming and its implications. Likewise, since matters related to material losses and substrates' fabrication constraints are increasingly severe at high frequencies, mmWave substrates need to be looked at; therefore, this paper details, as well, the particular issues related to mmWave propagation and manufacturing design processes for high-frequency devices. Aims at optimizing antenna and system reliability by employing advanced strategies and materials, as well as outlining the existing gaps that need clarification to augment 5G mmWave infrastructure and services.

Keywords: substrates; mmWave antenna design; 5G substrate materials; high-frequency dielectric characterization

1. Introduction

The coming of fifth-generation (5G) holds great potential in providing an ultra-fast data rate, a very low latency, and a significantly improved spectral efficiency by exploiting the millimeter-wave (mmWave) spectrum for the first time in mobile communication infrastructures as a requirement intended for a lot of emerging applications [1]. A number of studies [2] considered mmWave spectrum as a significant candidate for 5G services. At mmWave frequencies, systems are extremely influenced by the electromagnetic characteristics of substrate materials; therefore, accurate knowledge is vital for the finest and most robust design of such systems. The unique traits of mmWave communications pose a range of obstacles that limit their full deployment, spanning integrated circuit and overall system architecture, interference control, spatial reuse, obstruction compensation, and adaptive system oversight [3,4].

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Substrates, which can be either textured or uniformly flat, act as the foundational platform onto which conductive layers are subsequently applied [5]. The choice of substrate material is not merely a fabrication detail; it fundamentally influences device operation and end performance [5]. The authors of [6], for example, substituted a Rogers's substrate in an antenna design and observed pronounced performance variation, which they correlated to the substrate's dielectric constant and other mmWave parameters, a finding later validated in [7]. In selecting a substrate, engineers weigh a blend of criteria: electrical loss, machining precision, fabrication speed, and mechanical durability [8]. Consequently, substrate design must satisfy targeted performance metrics while also tolerating the chemical, thermal, and mechanical stresses imposed by the functional layers that are built on the top of them [9,10].

A number of mmWave studies have been identified in the literature [11]. These studies have addressed key issues pertaining to propagation behavior, applications, mmWave technologies, design considerations encompassing environmental conditions, usage scenarios, and operating frequencies [4,11,12]. Nonetheless, they do not offer an extensive examination of 5G mmWave substrates design [13]. Likewise, even with significant progress in materials and circuit technologies for 5G mmWave systems, several critical challenges remain insufficiently addressed in the literature [3,12,14].

One major gap is the lack of standardized dielectric characterization methodologies at frequencies above 60 GHz, where measurement uncertainties, fixture parasitics, and frequency dispersion complicate accurate extraction of permittivity and loss tangent [15–17]. In addition, the long-term reliability of emerging substrates, particularly flexible, polymer-based, and composite materials under thermal cycling, humidity exposure, and mechanical stress, is not yet fully understood, raising concerns about performance stability in real-world deployments [7,18,19]. Another unresolved issue relates to the scalable manufacturing of low-loss hybrid materials that can simultaneously satisfy electrical performance, mechanical robustness, and cost constraints for large-scale production [20–22]. These challenges highlight the need for a comprehensive evaluation of substrate technologies and design methodologies for mmWave systems.

For that reason, this paper aims to systematically review recent advances of 5G mmWave substrates design, challenges, and measurement aspects, analyze material and fabrication trade-offs, and identify research priorities to support the development of reliable, manufacturable, and high-performance substrates for 5G mmWave applications by targeting the most recent technical developments and analyzing potential upcoming research directions for the 5G mmWave substrates. Specifically, we focus on the following:

1. Recent development and research contributions towards mmWave communications.
2. The mmWave substrate design challenges and measurement approaches for 5G applications.
3. Substrate materials and the need for low dielectric constants and low loss tangents materials.
4. Fabrication methods and key design considerations for robust, high-performance mmWave devices.
5. Emerging technologies, including MIMO, AI-driven RSSI-aware beamforming and implications, and machine learning for optimization with their associated security concerns.
6. Significant recent research accelerating 5G mmWave substrate innovations.
7. Future research directions and identifying gaps in mmWave substrate development.

The remainder of this paper is structured as follows. Section 2 provides review methodology; Section 3 and its subsections provide an overview of mmWave frequencies and their significances, while Section 4 describes challenges of mmWave substrates. In Section 5, we discussed the fabrication techniques of mmWave substrates, and in Section 6,

we talk about thermal management in mmWave substrates. In Section 7, an overview of a substrate's electrical performance is provided. A description of the substrate's mechanical stability is given in Section 8. While Section 9 covers various hybrid mmWave substrate materials. Moreover, Section 10 provides a discussion about AI-driven RSSI-aware beamforming and its implications for substrate design, and finally, Sections 11 and 12 conclude the paper, which paves the path for future research, respectively.

2. Review Methodology

This review followed a systematic approach to identify, evaluate, and analyze recent studies related to mmWave antenna and substrate design for 5G. The process involved literature search, selection, data extraction, and normalization to ensure the inclusion of high-quality and relevant publications.

2.1. Literature Search and Databases

The literature search was conducted across several reputable scientific databases, including IEEE Explore, Science Direct, Springer Link, Scopus, and MDPI. Complementary searches were also performed using Google Scholar to capture additional relevant works not indexed in the above databases. The search strategy combined keywords such as "mmWave antenna design," "5G substrate materials," "graphene-based antennas," "SIW technology," 5, "MIMO", "AI-driven RSSI-aware beamforming", and "high-frequency dielectric characterization."

2.2. Publication Time Frame

To ensure relevance to current technological advancements, the publication window was limited and strictly to studies published between 2018 and 2026, covering the rapid development period of 5G and mmWave technologies. Older but highly cited benchmark studies were included only when they contributed foundational information on substrate behavior or design principles.

2.3. Inclusion and Exclusion Criteria

The inclusion criteria focused on peer-reviewed journal articles and conference papers, studies presenting experimental or simulated antenna/substrate performance above 24 GHz (mmWave range), research emphasizing substrate material properties (dielectric constant, loss tangent, and thermal stability), and works describing practical 5G applications, such as MIMO, SIW, AI-driven RSSI-aware beamforming, or graphene-based antennas. Exclusion criteria involved non-English publications, articles lacking quantitative performance data, patents, and unrelated low-frequency antenna and or substrate studies (<10 GHz).

2.4. Data Extraction and Normalization

Key information extracted from each paper included substrate type, dielectric constant (ε_r), loss tangent (tan δ), operating frequency, bandwidth, gain, and radiation efficiency. Data were normalized into consistent units (GHz, dBi, etc.) and categorized based on antenna type and application band. Where necessary, results were cross-checked across multiple sources to verify accuracy. The normalized dataset was tabulated to enable comparative analysis of substrate performance and to highlight materials demonstrating superior efficiency, stability, and suitability for mmWave and 5G applications. The flow diagram summarizing the review process is given in Figure 1.

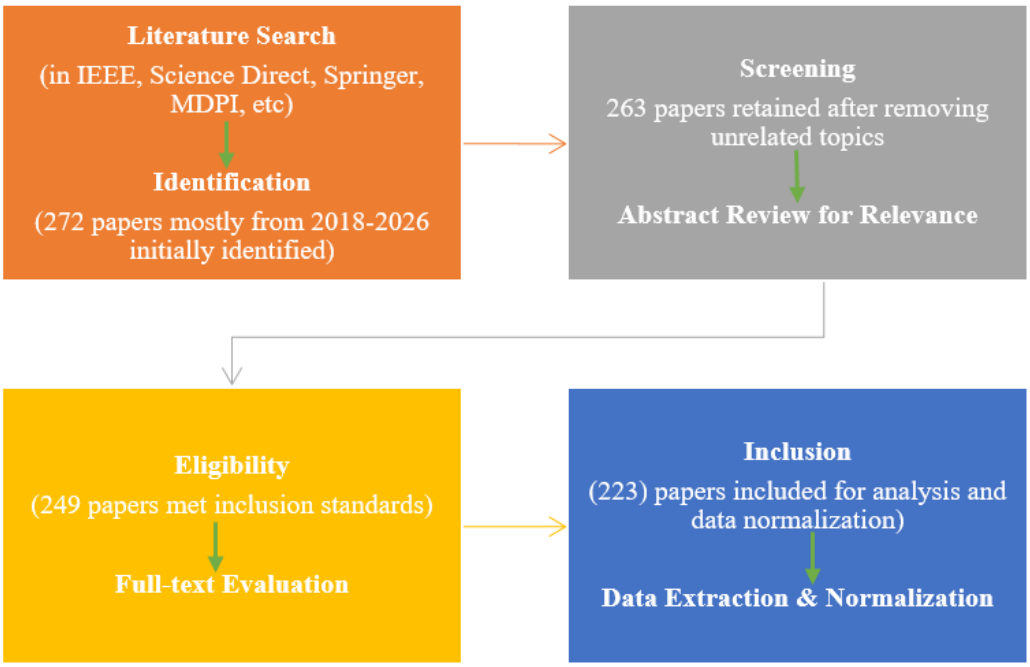


Figure 1. Flow of the review methodology.

3. Overview of 5G mmWave Frequencies and Their Significances

The mmWaves refer to frequencies ranging from 30 GHz to 300 GHz, corresponding to wavelengths between 1 and 10 mm [12,23]; see Table 1. Overview of the evolution of mobile communication generations from early cellular systems to 5G mmWave, emphasizing key technological transitions such as increased operating frequencies, adoption of beamforming, higher bandwidth requirements, and densification trends. As shown in Figure 2, the evolution of mobile communication systems toward mmWave operation introduces new performance requirements that directly influence substrate material selection and antenna design considerations. Thorough illustration of enabling technologies across mobile communication generations, clarifying the integration of phased arrays, RF front-end modules, and high-frequency interconnects. Similarly, Figure 2 emphasizes the relationship between system evolution and physical-layer constraints, including dielectric loss, thermal management, and fabrication considerations relevant to mmWave substrate design, discussed in this section as well as in Sections 4 and 6 below.

Table 1. The mmWave frequency bands [13].

Wave	Band	Frequency (GHz)	Wavelength (mm)
mmWave (30–300) GHz	Q	30–50	10–6
	U	40–60	7.5–5
	V	50–75	0.6–4
	E	60–90	5–3.33
	W	75–110	4–2.72
	F	90–140	3.33–2.14
	D	110–170	2.72–1.76
	G	110–300	2.14–1

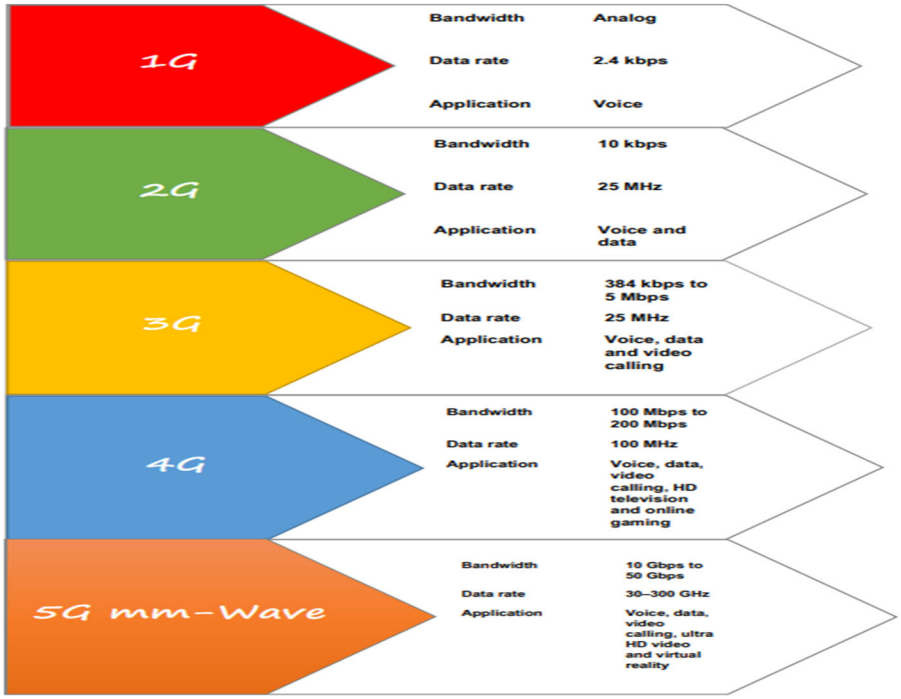


Figure 2. Mobile communication generations.

In the 5G context, mmWave commonly denotes the frequency band between 24 GHz and 100 GHz [24,25]. These frequencies offer very high bandwidth while supporting exceptionally fast data rates. However, signals at these frequencies are more prone to obstruction by buildings, vegetation, and rainfall, and are also more vulnerable to adverse weather due to their shorter wavelengths [1]. The mmWave bands are attracting significant research interest due to their potential to enable data rates approaching 10 Gbps, offer abundant bandwidth, support narrow beamforming, ensure high transmission quality, and provide strong detection capabilities [23].

The 5G spectrum in Figure 3 is structured into three distinct tiers that balance the laws of physics with the demands of modern data, beginning with the Low-band (<1 GHz), which serves as the foundational coverage layer by leveraging long wavelengths for deep indoor penetration and vast geographical reach. This is complemented by the Mid-band (1–6 GHz), frequently termed the "Sub-6 GHz sweet spot," where the network achieves an optimal compromise between reliable signal propagation and significantly increased data capacity. In contrast, the High-band or mmWave spectrum (>24 GHz) pushes into the 24 GHz to 100 GHz range to deliver ultra-high-speed, multi-Gbps throughput and near-instantaneous latency; however, because these high-frequency signals suffer from extreme path loss and atmospheric absorption [1], the deployment of sophisticated AI-driven beamforming becomes an absolute necessity to precisely focus and steer energy toward the user [26,27].

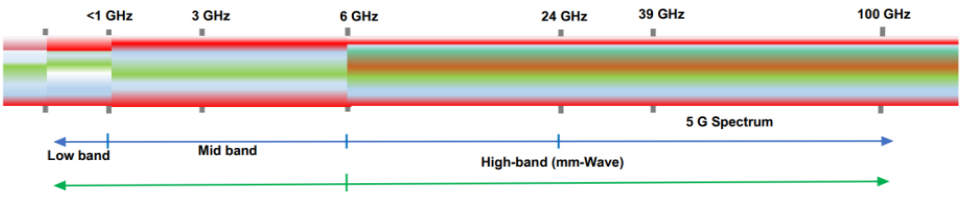


Figure 3. Comparison of the 5G NR spectrum allocation [3].

Precisely, Figures 2 and 3 highlight the transition from conventional sub-6 GHz systems to high-frequency mmWave architectures, emphasizing key technological drivers

such as beamforming, dense antenna arrays, and increased bandwidth demand, which directly motivate the need for advanced substrate materials and low-loss fabrication techniques discussed in later sections.

In recent years, mmWave frequency bands have attracted growing attention from academicians and researchers [4,12,28] due to their unique characteristics and a wide range of emerging applications, as illustrated in Figure 4.

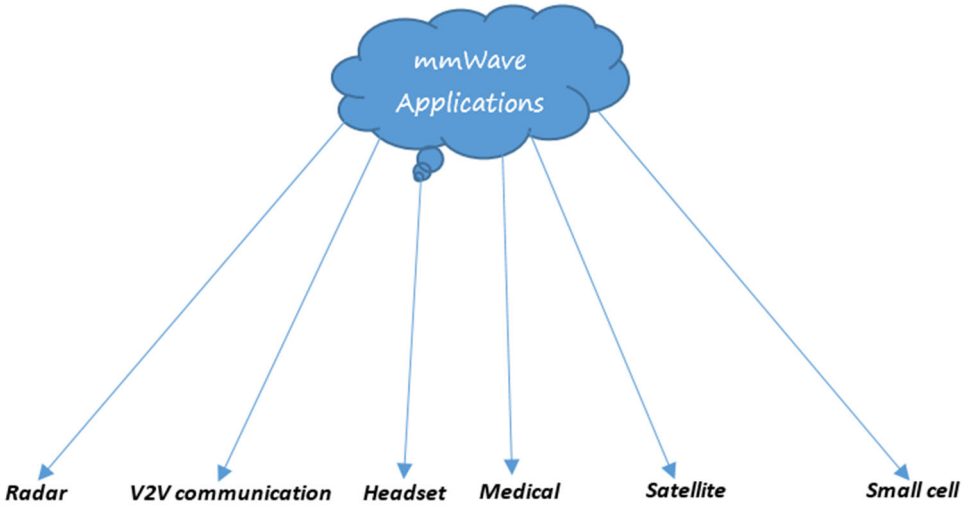


Figure 4. Deployment areas of mmWave frequencies.

3.1. 5G mmWave

At present, billions of wireless devices are engaged in global voice and data communication, leading to a remarkable surge in data traffic [29]. In response, 5G mmWave technology emerges with advanced features and enhanced services [30]. Due to the global bandwidth shortage, wireless carriers have turned to the mmWave spectrum, an underutilized frequency range in broadband communications, particularly the 24 GHz to 100 GHz band, now designed for 5G use, as can be seen in Figure 5. Currently, numerous studies and research efforts are available in the literature, focusing on mmWave propagation in both outdoor and indoor settings [11,28,31,32], particularly in densely populated areas [29]. The 5G mmWave is currently a less-utilized band; its signals transmit more data than lower frequency waves and can be integrated with MIMO antennas with the potential to offer a greater magnitude capacity compared to current communication systems [33].

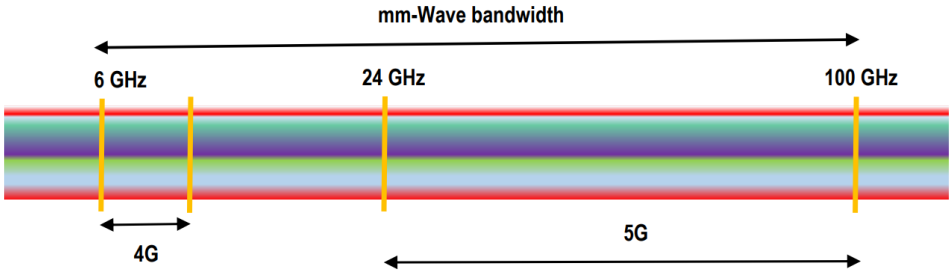


Figure 5. Illustration of mmWave frequency [3].

3.2. Advancements in mmWave Technologies

Recent advancements in wireless communication have been driven by the emergence of mmWave technology, which enables high processing speeds and increased integration density in modern electronic systems [1]. As illustrated in Figure 6, mmWave is being adopted in a range of current and emerging applications, such as IEEE 802.11ad WiGig, satellite communications, automotive radar, and airport security scanners [4,34].



Figure 6. The mmWaves advancements.

Over the last few years, there have been extensive studies aimed at developing mmWave antenna technologies to satisfy the performance requirements of 5G systems. Several works have proposed different designs targeting compactness, broad bandwidth, high gain, and MIMO compatibility. For example, ref. [35], in Figure 7, reported a 28 GHz MIMO antenna employing machine learning (ML) method to realize a 10.34 dBi gain and above 26 dB isolation, whereas the authors of [36] suggested an eight-port slot array for 5G mobile terminals as provided in Figure 8.

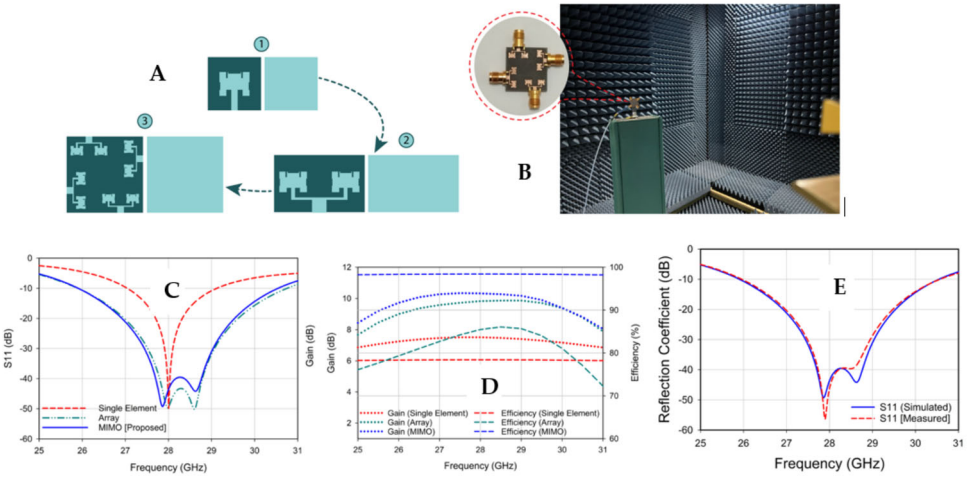


Figure 7. (**A**) General layout of the proposed antenna design: (1) single element, (2) antenna array, and (3) proposed MIMO configuration [35]. (**B**) Radiation pattern measurements conducted in an anechoic chamber. (**C**) Reflection coefficient results. (**D**) Gain and efficiency at each design stage and (**E**) reflection coefficient (S_{11}) of the proposed MIMO antenna [35].

Studies, such as [35], have contributed to the development of high-directivity, low-profile antennas with reconfigurable beamforming, circular polarization, and low mutual coupling, enhancing integration into dense and portable devices. Material investigations by the authors of [37] identified low-loss substrates suitable for mmWave ranges up to 115 GHz, while the authors of [38] proposed a compact metamaterial-based antenna for

26 GHz. Further, ref. [39] developed a square loop printed antenna with dual-beam support and high efficiency at 28 GHz; see Figure 9.

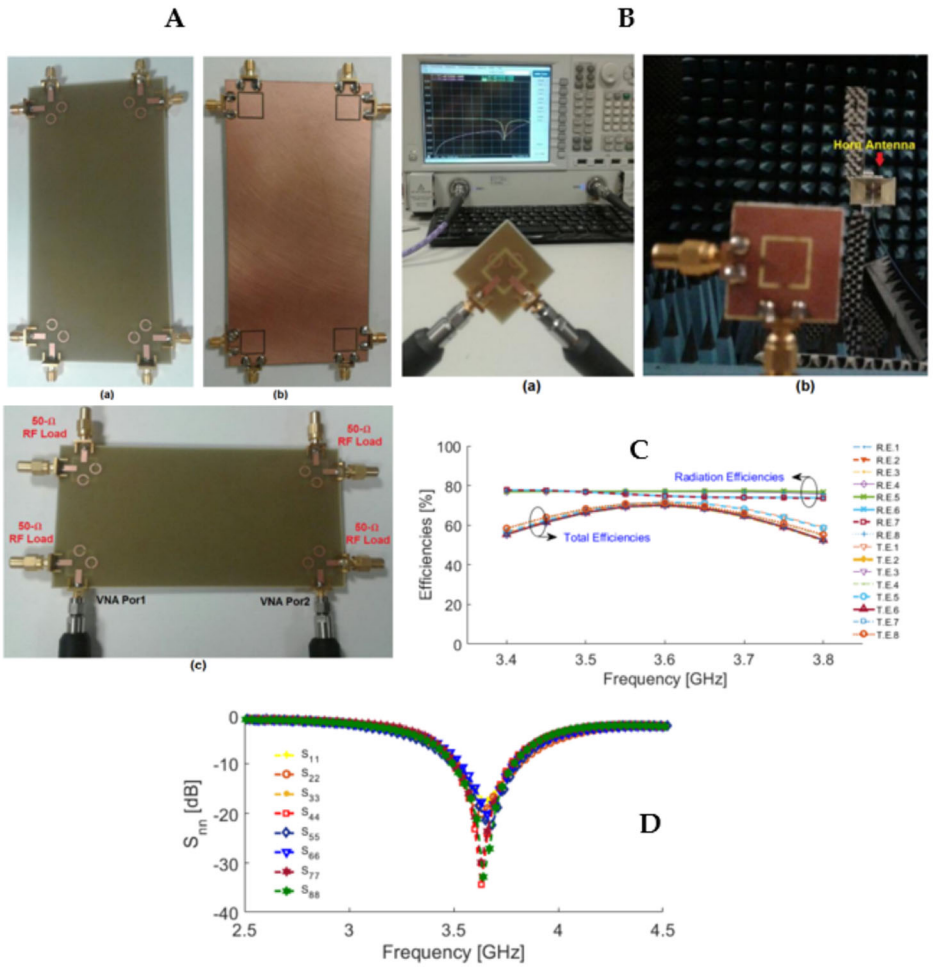


Figure 8. (**A**) Fabricated smartphone antenna PCB showing (**a**) top view, (**b**) bottom view, and (**c**) the prototype connected to cables and loads. (**B**) Measurement setups for (**a**) S-parameters and (**b**) radiation patterns of the fabricated antenna element [36]. (**C**) Antenna elements efficiency results. (**D**) Measured S-parameters results of the proposed smartphone MIMO antenna [36].

Directional horn antennas in [31] have enhanced beamforming capabilities and feeding networks for Ka-band applications. The work by the authors of [40] introduced a differentially fed broadband antenna array using multilayer PCB technology, optimized for base stations. The study in [1] contributed a graphene-based liquid antenna capable of beam reconfiguration across multiple 28 GHz frequencies. Additionally, ref. [41] demonstrated how 5G supports the integration of power IoT and cyber–physical systems in smart grids. Furthermore, ref. [42] developed an RSIW-based Antipodal Vivaldi Antenna with enhanced bandwidth, gain, and low signal loss, improving performance for 5G and radar systems. Likewise, ref. [43] designed a 50 µm polyimide on-chip SIW antenna, achieving compact size, low dielectric loss, and high efficiency for mmWave and IoT applications.

Together, these innovations advance compact, high-efficiency, and wideband solutions for next-generation mmWave technologies; they also collectively address key challenges in mmWave propagation, offering scalable and high-performance antenna solutions for next-generation 5G communication systems, as outlined in Table 2, in which a Cost Index (CI) on a 1–5 scale was used to compare mmWave designed device costs based on weighted factors of material price, fabrication complexity, and availability, offering a trans-

parent and consistent method to classify costs as low, moderate, or high without relying on variable market prices.

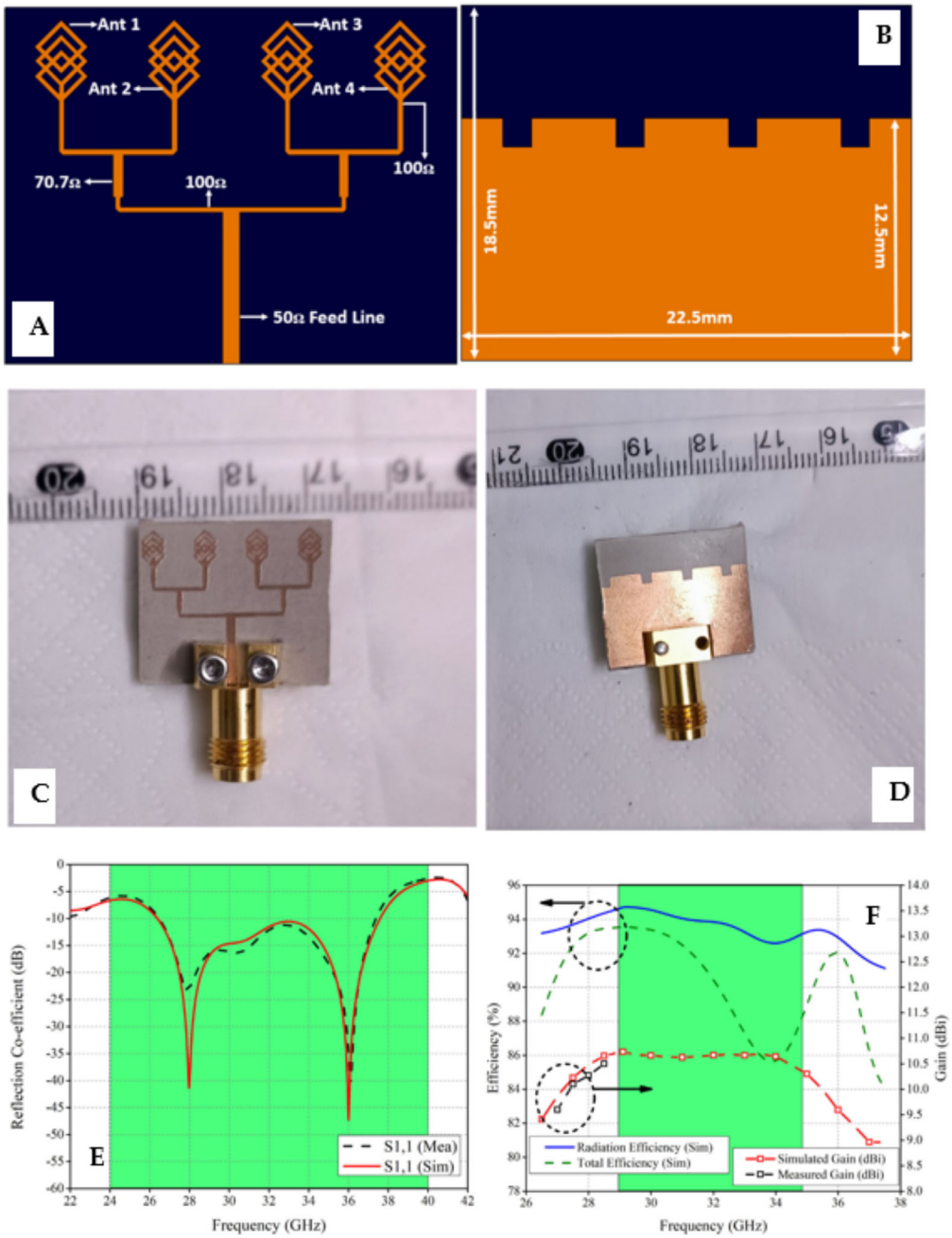


Figure 9. Configuration of the proposed antenna array: (**A**) front view and (**B**) back view. Fabricated prototype of the array: (**C**) front view and (**D**) back view. (**E**) Comparison of simulated and measured return loss. (**F**) Key performance metrics of the proposed array [39].

Table 2. Comparative analysis of mmWave antenna advancement technologies.

References	Designed Device	Relative Permittivity	Substrate Height (mm)	Gain (dBi)	Performance	Cost
[1]	Graphene liquid antenna	2.55	1	7	High gain, wide bandwidth for future 5G mmWave systems	Moderate
[29]	Dielectric resonator antenna	2.2	0.787	5.45	High efficiency and gain	Moderate

Table 2. *Cont.*

References	Designed Device	Relative Permittivity	Substrate Height (mm)	Gain (dBi)	Performance	Cost
[31]	Directional horn antenna	4.4	3	8.06	High directivity	Moderate
[35]	MIMO antenna integrated with ML based-gain prediction	2.2	0.787	10.34	Enhanced gain and bandwidth	High (ML integration)
[36]	8-element slot antenna with dual-polarization	4.4	1.6	3	Good isolation and efficiency	Moderate
[38]	Two-layer metamaterial antenna	2.2	0.78	11.27	High gain, wide bandwidth	High
[39]	Wideband three-loop antenna array	2.2	0.84	3.3	High isolation and ECC performance	Moderate
[42]	Antipodal Vivaldi antennas using RSIW	4.4	0.8	1.41	Improve the irradiation properties	Moderate
[43]	50 µm polyimide layer based on-chip antenna using SIW	3.5	0.1	1.5	Applicable to use in THz integrated circuits	Simple design process, and cost effective
[44]	8-element side-edge MIMO array	4.4	0.8	3.9	Enhanced isolation and gain	High
[45]	E-shaped H-slotted dual band mmWave MIMO antenna	2.2	0.8	7.9	Compact, good return loss	Low
[46]	Broadband dual-polarized MIMO antenna	4.4	1.6	4.8 and 5.8	High isolation, suitable for 5G	Moderate
[47]	4-port MIMO antenna	2.2	0.76	8.3	Compact, low mutual coupling	Low
[48]	3-dB hybrid coupler (printed-RGW)	2.94	0.762	3	High isolation, compact	Moderate

3.3. Substrates' Influence on mmWave Performance

Substrates provide the necessary physical environment and structural support for the circuits to operate effectively at mmWave frequencies. For the mmWave, substrates are important elements to support the signal propagation, thermal dissipation, mechanical reliability, component integration, and environmental stability, as well as the compatibility with the fabrication process [49]. Substrate technology is also essential for the implementation of the 30–300 GHz mmWave communication systems [50]. Substrate design generally involves the combination of different materials having different properties in order to achieve both good high-frequency electrical performance and good mechanical and environmental stability [50]. The role of substrates in mmWave technology is shown in Figure 10.

3.4. Types of Substrates Prevalently Used in 5G mmWave Systems

For 5G mmWave frequencies, substrate selection critically affects radiation efficiency, impedance stability, and thermal performance. Low-loss laminates and LCP-based hybrids are commonly adopted in practical 5G antenna implementations due to their favorable dielectric loss and manufacturability, while ceramics provide superior thermal and dimensional stability for high-reliability systems [20,51–53]. Conventional organic substrates such as FR-4 exhibit excessive dielectric loss and are generally unsuitable for high-efficiency mmWave radiators [54]. The comparison of typical substrate types for mmWave

applications in Table 3 highlights the trade-offs among electrical performance, thermal management, manufacturability, and cost when selecting substrates for mmWave systems. Laminates and hybrid materials offer a practical balance for 5G applications, while ceramics are preferred in high-power or high-reliability environments, and organic substrates remain suitable for cost-sensitive designs.

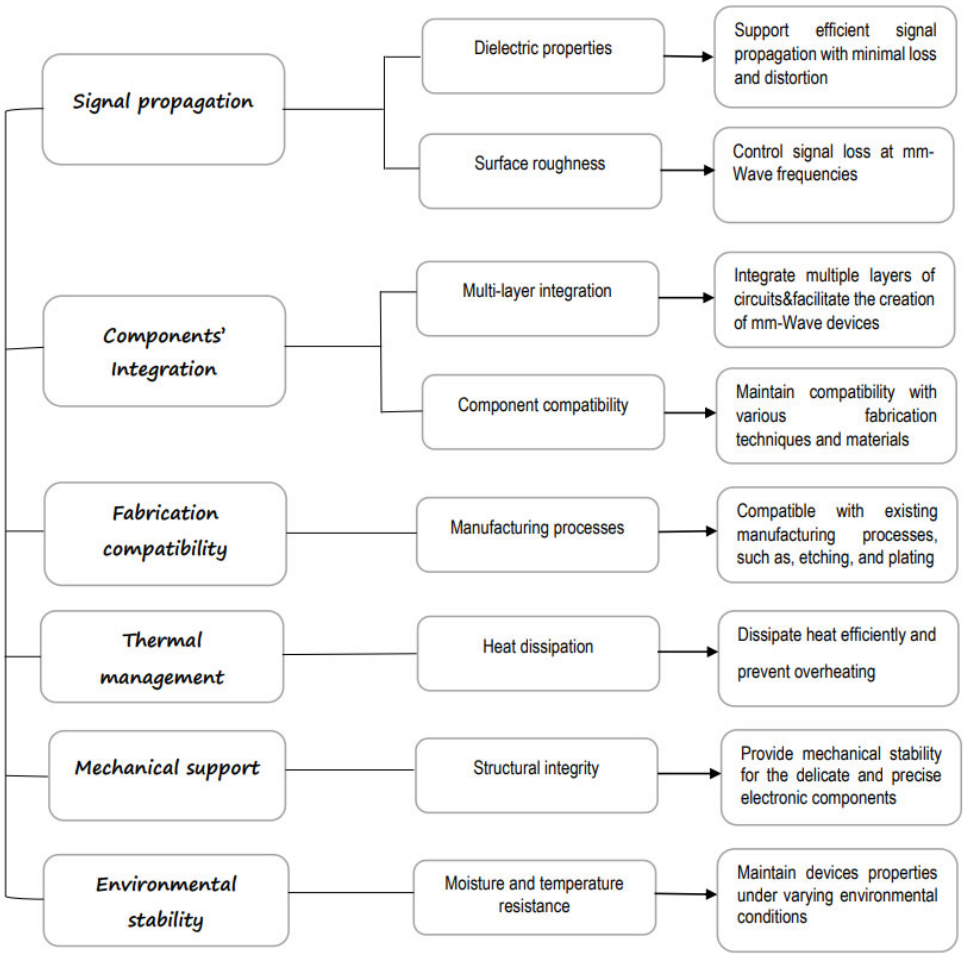


Figure 10. Role of substrates in mmWave technology.

Table 3. Comparison of typical substrate types for mmWave applications.

Substrate Type	Typical Materials	Dielectric Constant	Loss Tangent	Thermal Conductivity	Cost Index *	Key Advantages	Limitations	Application Scenarios	Ref.
Laminates	PTFE-based (e.g., Rogers), hydrocarbon/ceramic laminates	2.2–3.5	0.0007–0.003	0.2–0.6	Medium–High	Low loss, stable dielectric properties, mature PCB processing	Higher cost than standard PCB, limited thermal handling	5G antennas, phased arrays, RF front-ends	[17]
Organic	FR-4, polyimide, epoxy composites	3.5–4.7	0.01–0.025	0.2–0.4	Low	Low cost, easy processing, widely available	High loss at mmWave, poor dielectric stability	Low-cost RF boards, prototypes, sub-6 GHz systems	[54]
Ceramics	Alumina, LTCC, HTCC	5–10	0.0005–0.002	3–25	High	Excellent thermal stability, high power handling, low loss	Brittle, expensive, complex fabrication	High-frequency modules, radar, space electronics	[55]
Hybrid	Ceramic-filled polymers, LCP, glass-reinforced composites	2.5–4	0.001–0.005	0.3–1.5	Medium	Balance of performance and cost, tunable properties	Processing complexity, material variability	mmWave packages, antenna-in-package, flexible RF	[56]

* Cost Index (CI): a relative comparison on a 1–5 scale was used to compare typical substrate types for mmWave applications based on weighted factors of material price, fabrication complexity, and availability, offering a transparent and consistent method to classify costs as low/medium/high without relying on variable market prices.

To strengthen the quantitative assessment, standard value ranges for dielectric constant, loss tangent, and thermal conductivity across common mmWave substrate classes and operating bands (28, 39, and 60 GHz) are incorporated and linked to the comparative analysis provided in Table 3. These ranges are consistent with reported measurements in recent studies on low-loss laminates, ceramics, glass substrates, and hybrid materials used in phased arrays and high-frequency interconnects [17,20,22,51–53,55–57]. The inclusion of frequency-dependent property ranges enables clearer benchmarking of material trade-offs and supports more rigorous evaluation of substrate suitability for high-efficiency antenna systems, as well as strengthening the cost–performance presented in Table 3.

Table 4 provides a cross-referenced substrate classification to ensure consistency with Table 3 and provides frequency-dependent benchmarks for evaluating dielectric and thermal performance across different substrate classes. According to the data given in Table 4, laminates offer low permittivity suitable for wideband antennas [17,51,58–60], and ceramics provide superior thermal performance and dimensional stability, illustrating inherent trade-offs that must be considered in mmWave system design [53,55,61–63], since at mmWave frequencies (28–60 GHz), dielectric dispersion becomes more pronounced due to polarization relaxation and increased dipolar loss mechanisms, leading to slight increases in effective permittivity variation and measurable rises in loss tangent with frequency [64]. Materials such as PTFE-based laminates exhibit relatively stable ε_r across this band [17,19,51,65], whereas organic substrates and some hybrid systems show stronger frequency dependence due to molecular polarization effects [64]. This trend highlights the importance of frequency-specific characterization when designing antenna substrates for broadband 5G applications.

Table 4. Characteristics ranges of typical mmWave substrates (28–60 GHz).

Substrate Type	ε_r			tan δ			Thermal Conductivity (W/mK)	Remarks	Ref.
	28 GHz	39 GHz	60 GHz	28 GHz	39 GHz	60 GHz			
Low-Loss Laminates	2.1–3.2	2.1–3.3	2.0–3.4	0.0005–0.002	0.0007–0.0025	0.001–0.003	0.3–0.6	Widely used in 5G arrays	[17]
Glass/Quartz	3.8–4.8	3.8–4.8	3.7–5.0	0.0002–0.001	0.0003–0.001	0.0005–0.002	1–2	Low loss, rigid	[22]
Organic Substrates	2.9–3.4	2.8–3.5	2.8–3.6	0.002–0.005	0.003–0.006	0.004–0.008	0.2–0.4	Flexible electronics	[54]
Ceramics	5–9	5–9	5–10	0.0001–0.0008	0.0002–0.001	0.0003–0.0015	3–25	High stability, high cost	[55]
Hybrid Composites	2.5–4.5	2.5–4.7	2.4–4.8	0.0008–0.003	0.001–0.004	0.0015–0.005	0.5–5	Tunable performance	[56]

According to the authors of [66], substrate divides into discrete and integrated. Discrete substrates are usually insulators, while integrated substrates are, of course, semiconductors [50]. Discrete substrates, therefore, have a lower dielectric constant than integrated substrates, providing better electrical properties for passive components, especially in the RF and mmWave domains [66]. Various substrate fabrication methods yield substrates with distinct characteristics, and they are typically categorized based on the materials employed and the techniques used in their production [50]. Despite significant early advancements and innovations, the field of 5G mmWave substrates still faces numerous challenges before achieving widespread practical adoption, presenting substantial opportunities for further scientific research and development [49]. Research on mmWave substrates for 5G has grown gradually since 2015, as shown in Figure 11, indicating increasing academic awareness and technological significance.

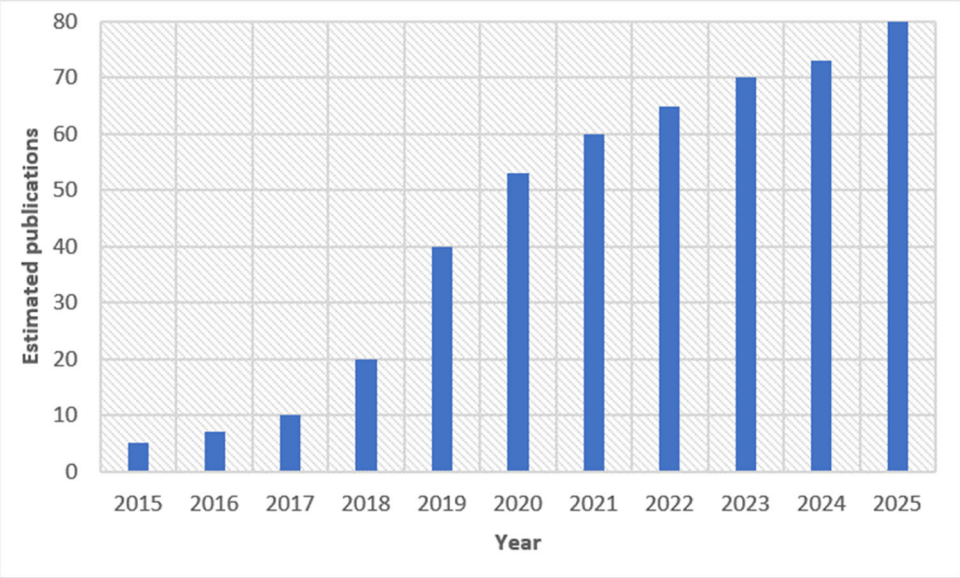


Figure 11. Projected number of research papers per annum for "5G mmWave substrates". Source: Web of Science trends.

3.4.1. Laminate Substrates

Laminate substrates are essential in mmWave, as their dielectric characteristics are of paramount importance during the layout for circuit and subsystem design [67]. However, this type of substrates suffer from cost inefficiency, with negative trade-offs for both materials and fabrication [50]. Full spectrum-based measurements on several of these materials have been performed using different techniques and spanning from the low to the very high frequency, providing valuable data for engineers designing up to THz frequencies [67]. There are two typical laminate substrates used for the fabrication: FR-4 and Duroid. The properties of Duroid, such as efficient uniform transmission of high-frequency signals and negligible signal loss, slight thermal problems (thermal management challenges associated with heat generation, temperature rise, and thermally induced performance degradation in high-frequency substrates), make it an ideal candidate in new generations of communication devices. The study in [68] introduced a design strategy of low-loss, low-permittivity dielectric laminate substrates for microwave and mmWave frequency systems using mixing theory.

3.4.2. Ceramics Substrates

Ceramics are very attractive substrates for low-loss mmWave frequencies and are seeing increasing usage due to their appealing features. For example, Alumina Ribbon Ceramic (ARC) has been reported for 5G mmWave patch antennas that have low loss and a small design applicable for mobile communication [69]. The authors of [70] presented the dielectric properties of a novel ultra-low-temperature co-fired glass ceramic composite proposed as a dielectric substrate for mmWave applications. Advantages of the ceramic substrates are stability, low signal loss, accurate line widths, high thermal conductivity, and both heat and cold resistance. Usually, they are characterized as thin-film, thick-film, low-temperature co-fired, and high-temperature co-fired ceramics [50].

3.4.3. Organic-Based Substrate Materials

Organic substrates are fundamentally lamina formed of organic-based composite layers [66]. On the other hand, they show favorable electrical properties and low-cost fabrication but are considered to be an enormous challenge in dimensional precision and surface roughness [8]. A liquid crystal polymer (LCP) is a well-known example of organic lam-

inates with excellent properties at mmWave [18]. For example, the dielectric constants of LCP materials fall in the 2.0–4.5 range at frequencies within 140–220 GHz, and some exhibit nondispersive performance [71]. For example, the authors of [72] designed a conformal ultra-wideband MIMO antenna on LCP substrate for 5G and mmWave IoT applications. Cyclicolefinpolymers (COPs) and polymer-ceramic composites are other organic substrates [66]. Likewise, the authors of [73] proposed an innovative multicore PCB and organic substrate method for the realization of the broadband dual-polarized antennas at 28 GHz and 39 GHz intended for 5G mmWave communication.

3.4.4. Other Materials Substrates

At mmWave frequencies, thin-film substrates such as glass or sapphire, hybrid composite, and flexible substrates can be used [50]. Glass, mainly composed of amorphous silica (SiO_2), exhibits a low loss tangent and low coefficient of thermal (CTE), making it a suitable substrate for high-frequency applications. Sapphire, which is a single-crystal form of Al_2O_3 with a hexagonal crystal structure, has similar advantages to alumina but with better surface smoothness, in many cases rendering it the ideal choice. In Figure 12, the main classes of substrates appropriate for mmWave communication are presented.

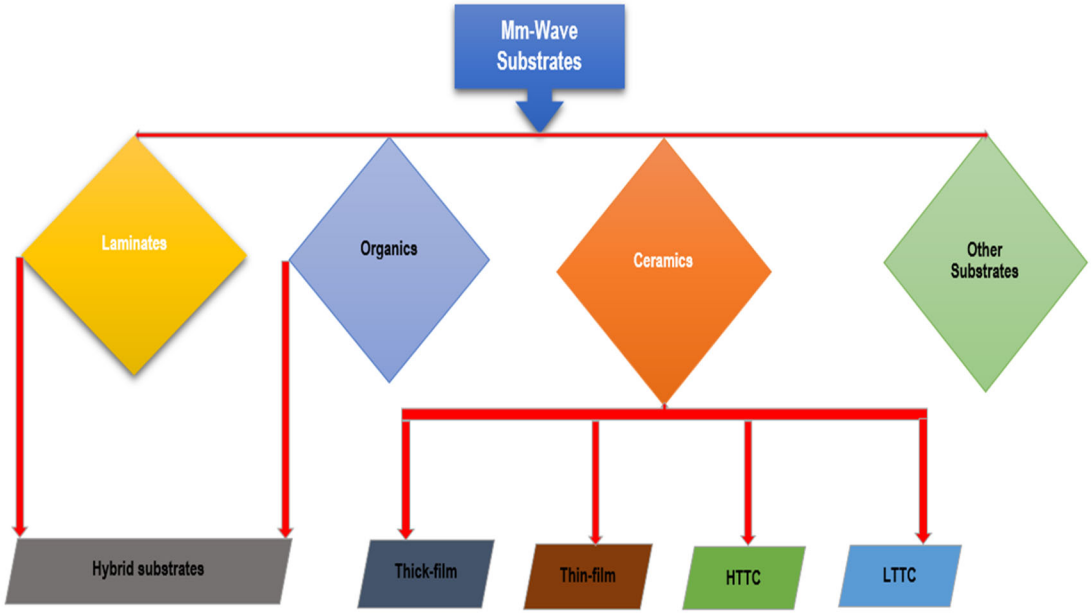


Figure 12. Type of substrates used in mmWave communication.

3.5. Key Properties and Requirements for mmWave Substrates

Selecting a substrate for mmWave applications is currently a major challenge for academicians and researchers. For instance, the authors of [6] extensively investigated how different substrate dielectric materials affect the performance of a wideband magneto-electric dipole (MED) phased array antenna, utilizing four commercially available substrates for mmWave applications. Their study offers a foundational insight into how dielectric properties influence substrate performance and serves as a valuable reference for selecting suitable dielectric substrates for 5G mmWave systems. In order to attain high-speed data transfer from device to device with minimal error, employing materials of low dielectric constant (ε_r) and dissipation factor (tan δ) should be regarded as a main factor. Recently, numerous low ε_r and tan δ materials have been suggested, for example, modified polyimide (MPI), epoxy, polyphenylene ether (PPE), and polybenzoxazole (PBO). Nevertheless, the dielectric properties and moisture stability of the aforementioned materials are inadequate for high-speed applications [56]. Therefore, achieving high signal integrity

and low transmission delay at high-frequencies has directed academic and research efforts toward developing interlayer dielectric materials with low ε_r and tan δ [19].

Similarly, to construct mmWave systems effectively, the substrate material should possess a low ε_r [20], as a lower ε_r reduces capacitance, minimizing crosstalk from capacitive coupling [20]. It also facilitates faster signal transmission through the substrate, enabling higher data rates and reduced latency [37]. Additionally, a low tan δ ensures minimal signal attenuation in passive components and supports highly efficient antenna performance [32]. These low-loss dielectrics allow to integrate high-quality off-chip elements like inductors, filters, resonator in the first redistribution layer (RDL) [74]. In addition, the low tan δ values reduce the inherent high attenuation at mmWave frequencies, leading to the acceptable signal transmission inside devices [37]. These low-loss dielectrics facilitate integration of high-quality off-chip components, such as inductors, filters, and resonators in the first redistribution layer (RDL). Furthermore, by lowering the tan δ values, any naturally high attenuation at mmWave frequencies can also be suppressed to ensure sufficient transmission of the signal in the devices.

3.5.1. Dielectric Constant

The dielectric constant (ε_r), commonly known as the relative permittivity, describes a material's atomic-level response to an electric field and is normally in the range 3–4 for discrete substrates [50]. With a growing tendency, ε_r' s role is also growing in the behavior of materials [50]. Materials with lower dielectric constants allow for faster signal propagation and reduced attenuation [50]. Thus, in order to achieve an efficient mmWave operation, a substrate should be homogeneous in terms of dielectric characteristics throughout its body [50]. Different studies have explored the influence of ε_r on device performance for 5G mmWave applications [75]. The work by the authors of [76] investigated the influence of substrate ε_r and thickness for 5G application on a 38 GHz rectangular microstrip antenna. Similarly, ref. [77] analyzed a microstrip patch antenna at 28 GHz with various dielectric substrates. Additionally, ref. [37] introduced a full characterization of dielectrics for the selection of 5G mmWave candidates.

3.5.2. Loss Tangent (Dissipation Factor)

The loss tangent (tan δ) or dissipation factor quantifies dielectric losses in substrate materials, and it is another important factor besides ε_r [50]. Reduced tan δ helps to lower the attenuation in transmission lines for better-performing signals [50]. In addition, the quality factor of the dielectric is the reciprocal of tan δ that indicates its efficiency at high frequency [50]. The authors of [78] investigated the effects of tan δ and conductive surface roughness in transmission line losses at 5G mmWave frequency. They showed that conductor loss is dominant in a thin substrate and also indicated the advantages of low-roughness copper foil. The entire dielectric characterization of several materials, 5–115 GHz, showed ε_r and tan δ to be approximately constant, such that it grows linearly with frequency [37]. There are many studies on the impact and improvement of tan δ in substrate materials for 5G mmWave applications.

For instance, ref. [65] studied the influence of dielectric properties on the performance of a 5G planar branch line coupler. Ref. [21] also reported a new low-loss dielectric material at 10 GHz, as this frequency is widely used in material datasheets and standardized measurement techniques, allowing consistent comparison across different substrate technologies. Furthermore, ε_r and tan δ exhibit frequency dispersion; the general material trends remain indicative of performance at higher frequencies. In addition, at mmWave frequencies, dielectric losses generally increase due to dipolar relaxation, surface roughness effects, and increased conductor loss contributions. Hence, extrapolation from 10 GHz

data should be interpreted cautiously, and an accurate mmWave design requires characterization closer to the intended operating band or the use of frequency-dependent models. Likewise, ref. [79] created a low tan δ adhesive for reducing transmission losses in mmWave substrates. Equally, ref. [80] reported a newly developed dielectric material with tan $\delta < 0.003$ at 39 GHz, applicable in 5G mmWave RF modules.

3.5.3. Electrical Conductivity

Electrical conductivity (σ), the reciprocal of resistivity, is a consequence of free electrons in the atomic structure of a material [50]. In substrate design, conductivity is highly required, especially in metallic layers, but also significantly affects semiconductor behavior [50]. There has been extensive research aimed at σ measurement in 5G mmWave substrates. The authors of [81] developed a broadband measurement approach using a balanced-type circular disk resonator (BCDR) for assessing σ and ε_r from 20 GHz to over 100 GHz. Their findings indicate that as temperature and frequency increase, substrate tan δ rises while σ decreases. Ref. [78] examined loss properties of different transmission lines; they discovered that conductor loss dominates in thin substrates, and low-roughness copper foil is effective in reducing losses. The authors of [82] measured the electrical parameters, including σ of materials used as substrates for wearable antennas in 5G frequency bands. Ref. [83] introduced a low-loss dielectric material for 5G RF modules, demonstrating superior electrical reliability and filter performance in the mmWave frequency range.

3.5.4. Skin (Penetration) Depth

The skin depth is a frequency-dependent property that significantly affects the selection of metallization in substrates. It refers to the phenomenon where, at high frequencies, RF currents are confined primarily to the surface layer of a conductor, rather than flowing through its entire cross-section [50]. The study in [84] presented a scheme to design an electromagnetic wave (EM) absorber based on Ni deposition that forms a sub-skin depth periodic pattern, which induces resistive (rather than reflective) properties on the structure, resulting in enhanced absorption of EM waves.

3.5.5. Thermal (Heat) Properties

Thermal properties are vital to the design of mmWave substrates since heat density increases with frequency. Important thermal properties are thermal conductivity (the material's ability to conduct heat, often large in materials with a lot of free electrons, e.g., diamond, TPG, and CNTs) and thermal expansion, which is measured by the thermal expansion coefficient (CTE) [50]. To inhibit thermal mismatch, one would design effective substrates with low CTE or with compatible component materials. With low CTE, materials such as laminates and TPG [50] are recommended for high-frequency applications. Studies recently performed have carried out a thorough investigation on the thermal properties of mmWave substrates, focusing on their reliability and conductance capability at different thermally disparate environments. The authors of [52] have shown that the hybrid substrates outperform the FR-4 in terms of thermal reliability for mmWave modules. Another paper in [85] demonstrated that β-Si_3N_4 is an excellent substrate for heat dissipation with decent thermal conductivity as well as good dielectric properties for wireless communication. The authors of [86] have also shown the influences of loss tangents and conductivity on the propagation of microstrip lines with the increase in temperature and frequency. Additionally, the authors of [68] designed a low-loss, low-permittivity dielectric substrate with superior electrical performance on Maxwell Garnett's mixing theory for high-frequency-related applications.

3.5.6. Dielectric Breakdown (Insulation Failure)

Dielectric breakdown is a critical property of substrate materials, defined as the voltage threshold at which an insulating material begins to conduct electricity due to excessive electric field strength. When this threshold is exceeded, current flows through the dielectric, compromising its insulation capability [50]. In the context of mmWave 5G systems, the authors of [6] studied how dielectric substrates influence the performance of magneto-electric dipole phased array antennas, while the authors of [67,76] characterized the dielectric behavior of common substrates to evaluate their suitability for high-frequency applications.

3.5.7. Reliability and Mechanical Stress Factors

The reliability of the electronic system generally refers to its ability to function consistently over time without failure [50]. In the context of substrates, reliability is more specifically associated with a material's capacity to withstand stress and strain. Stress reflects internal forces, while strain results from external loads. Young's modulus, defined as the ratio of the tensile stress to the tensile strain, quantifies a material's resistance to deformation [50]. The study in [87] investigated the reliability of mmWave modules under thermal cycling, linking solder joint failures to substrate warpage. Later, the authors of [52] found that mmWave modules built on hybrid substrates showed superior reliability under temperature stress compared to those on conventional FR4 substrates.

3.5.8. Additional Substrate Characteristics

Other important substrate characteristics include thickness, glass transition temperature, decomposition temperature, flammability, moisture absorption, chemical resistance, and surface roughness [50]. As shown in [88], glass substrates demonstrated low dielectric loss and smooth surfaces, making them promising candidates for 5G and mmWave applications.

3.6. Advancements in mmWave Substrates

Recent advancements in mmWave substrates show a clear move toward tunable and high-performance materials. Currently, a wide range of mmWave devices have been developed using different substrate materials to meet the demands of 5G applications [89]. Several studies, such as those by the authors of [54], explored microscript patch antennas fabricated on Rogers RT/Duroid-5880, FR4-epoxy, and Rogers RT5880, respectively, highlighting both low-cost and high-performance options based on their varying ε_r and tan δ. Moreover, studies by the authors of [90] proposed an array of MIMO antennas with various substrates, such as FR-4, Rogers RT5880, RO5880, Duroid-5880, and RO4350B, to achieve a high gain, isolation, and broad bandwidth. These works affirm that, by proper selection of substrate and optimized antenna geometries, a compact and high-gain 5G mmWave system is realized. In addition, ref. [91] further verified the favorable performance of Rogers RT5880 for low-loss antenna designs, and the authors of [53] proposed the employment of barium fluoride (BaF_2) ceramics to mmWave applications based on LTCC technology.

Similarly, the study in [1] introduced a graphene-based liquid antenna on an LCP substrate, achieving 360° beam reconfiguration and about 7 dBi gain at 28 GHz, highlighting flexibility and reconfigurability. The authors of [92] also used a bilayer graphene film substrate enabling frequency tuning between 29.6 and 40 GHz with low reflection losses, proving graphene's potential for adaptive mmWave systems. Meanwhile, ref. [35] utilized a Rogers RT/Duroid 5880 substrate for a 28/38 GHz MIMO antenna, achieving high isolation and wide bandwidth. Together, these studies demonstrate that combining traditional

low-loss substrates with emerging graphene-based materials enhances mmWave antenna performance, tunability, and efficiency for 5G applications.

Furthermore, ref. [56] developed advanced thermosetting siloxane-hybrid substrates with low dielectric constant, low loss tangent, and high stability under humid conditions, enhancing performance and reliability for mmWave microstrip lines and filters. Meanwhile, ref. [93] focused on accurate dielectric characterization at THz frequencies, enabling precise evaluation of substrate permittivity and losses using THz measurement techniques. Together, these studies advance mmWave technology by improving substrate material quality and providing accurate data essential for efficient high-frequency antenna and circuit design. The studies in [94,95] demonstrate significant advancements in mmWave substrate applications for MIMO antenna systems by designing a 5 × 20 massive MIMO antenna using a low-loss Rogers RT/Duroid 5880 substrate, achieving high gain, wide bandwidth, and excellent isolation, key for 5G base stations, and they also developed a 2 × 2 MIMO antenna array optimized on a low-dielectric, high-stability substrate, delivering improved radiation efficiency, compactness, and reduced mutual coupling, respectively. Collectively, both studies highlight how selecting low-permittivity, low-loss substrates greatly enhances MIMO antenna performance, enabling efficient, high-speed mmWave communication

The results carried out by the above authors demonstrate that a significant progress has been made in 5G mmWave antenna design and optimization of the substrate material. All mentioned substrates were selected based on the desired electrical and mechanical characteristics, and so, for instance, FR-4 was found to be preferable for an inexpensive, small-size design with moderate performance and extreme Rogers and Duroid in the case of supporting high frequency application with a low loss, a high gain, and a wide bandwidth. Furthermore, BaF_2 ceramics have been designed for LTCC applications with higher Q values, lower sintering, and better bonding to metal and polymer. Moreover, for 5G and mmWave applications, substrates with low dielectric constant and minimal loss are ideal. Rogers RT/Duroid 5880, RO4350B, Arlon Diclad 880/527, and CLTE-MW offer excellent performance due to their low dielectric permittivity (≈2.1–3.5), low loss tangent (≈0.0009–0.003), and high thermal stability, making them highly efficient at 26–60 GHz. Graphene improves the advantages of tunable conductivity and flexibility for reconfigurable antennas, while BaF_2 provides very low permittivity and loss, suitable for ultra-high-frequency designs. In contrast, FR-4 and FR-4 epoxy, with high losses and unstable dielectric behavior, are unsuitable for mmWave systems. Overall, Rogers-, Arlon-, CLTE-MW-, and graphene-based substrates are the most suitable for high-performance 5G antenna designs. A summary of advances in mmWave substrates in antenna applications is given in Table 5.

Table 5. Summary and progress of 5G mmWave antenna developments.

Reference	Substrates	mmWave Device	Frequency	Device Performance
[1]	Graphene	Beam-reconfigurable graphene liquid antenna	28 GHz	Performs beam-steering up to 360° with 7 dBi of gain and a wideband of 10 dB impedance bandwidth of over 20% for 5G mmWave wireless communication systems
[35]	Graphene	28 GHz MIMO Antenna for 5G applications	28	Enhanced due to tunable graphene conductivity
[53]	BaF_2	mmWave antennas	mmWave range	High reliability, thermal stability and low loss
[54]	FR-4 epoxy	Compact microstrip patch antenna	40 GHz	Compact design, moderate gain and cost-effective
[56]	CLTE-MW	Microstrip line and filter structures	60	Stability maintained under high humidity; low ε_r/tan δ improves frequency stability

Table 5. *Cont.*

Reference	Substrates	mmWave Device	Frequency	Device Performance
[90]	FR-4	Compact multiband MIMO antenna system	Compatible with 5G/WLAN/WiFi-6 standards	Multiband, high diversity, and compact design
[91]	FR-4	C-slot microstrip patch antenna	30 GHz	Good return loss, simple structure, and low-cost material
[92]	Graphene	Microstrip patch antenna	29.6–40 GHz	The operating mode of the antenna achieved frequency reconfigurability
[93]	CLTE-MW	Dielectric terahertz characterization	sub-THz and THz	Revealed a more pronounced frequency-dispersive behavior among different materials
[95]	Arlon Diclad 880	5 × 20 massive MIMO antennas	28.1	Best suited for the development of a microstrip patch antenna array for the 5G wireless handheld communication devices
[94]	Arlon Diclad 527	2 × 2 MIMO antenna array	28	High isolation due to low-loss substrate and optimized spacing
[96]	Rogers RT5880 (lossy)	Elliptical slot circular patch antenna	Dual-band operation at 28 GHz and 38 GHz	Supports dual-band operation, optimized for 5G mobile applications with high isolation
[97]	Rogers RT/Duroid-5880	Rectangular microstrip patch antenna	28 GHz	Low profile, high gain, and suitable for compact devices
[98]	FR-4	Wideband MIMO antenna with symmetry	mmWave band	Polarization diversity and compact design
[99]	FR-4	Two-monopole MIMO with parasitic elements	Sub-6 GHz	High isolation and improved MIMO performance
[100]	FR-4	Super-wideband MIMO antenna	2.3–23 GHz	High isolation and optimized bandwidth
[101]	FR-4	4-Port octagonal MIMO antenna	3.5–11 GHz	Low mutual coupling and broadband
[102]	Rogers RT/Duroid 5880	Dual band high gain MIMO antenna	28/37 GHz	High gain and compact 4-port design
[103]	FR-4	UWB MIMO with X-shaped stubs	1.9–14 GHz	High isolation and good bandwidth
[104]	Rogers-RO4350B	High Gain Array MIMO antenna	28 GHz	High gain, array performance, for mmWave

4. Challenges in mmWave Substrates Development

The advancement of substrate technologies for 5G and mmWave systems is constrained by a combination of electromagnetic, material, manufacturing, and economic challenges [3,105]. At frequencies above 24 GHz, signal propagation becomes highly sensitive to dielectric loss, conductor surface roughness, and dimensional tolerances, requiring substrates with tightly controlled properties [49,105]. While significant progress has been achieved through low-loss polymers [19], ceramics [53], and hybrid composites [56], several unresolved issues remain that limit performance optimization and large-scale deployment.

4.1. Technical Challenges

From a technical perspective, signal loss represents one of the most critical challenges. Both dielectric loss and conductor loss increase with frequency due to polarization relaxation and reduced skin depth, leading to attenuation and reduced antenna efficiency [16,78]. Current approaches focus on low-loss materials such as PTFE-based laminates, liquid crystal polymer, and ceramic substrates; however, these materials often involve trade-offs in mechanical robustness, processing complexity, or cost [19,21]. Additionally, surface roughness effects become increasingly dominant at mmWave frequencies, necessitating smoother conductors or advanced roughness correction models, yet these solutions may compromise adhesion or increase fabrication difficulty [78].

Accurate dielectric characterization at mmWave frequencies remains another major challenge [67,106]. Measurement techniques such as resonator methods, transmission line

extraction, and free-space characterization can produce inconsistent results due to calibration sensitivity and fixture dependence [78,107,108]. The lack of standardized testing protocols above 60 GHz complicates material comparison and design validation, highlighting the need for improved measurement frameworks [23]. In mmWave system design, the use of substrates with low ε_r and tan δ is essential, if one wants to obtain broader bandwidth and lower signal loss [13]. Despite these many benefits, mmWave communication remains troubled with certain technical issues such as high pass loss, atmospheric attenuation, signal blockage, interference, and lack of spatial reuse [3].

Non-technical obstacles are the high cost of infrastructure and low user digital literacy [3,12]. The study by the authors of [105] highlighted issues like narrow beamwidth and high directivity due to the short wavelengths of mmWaves. Accurate channel modeling is necessary for effective deployment, as signal generation, reception, and propagation remain difficult at such high frequencies [109]. To address these, ref. [110] recommended using the directional antennas with enhanced gain at both transmitter and receiver ends.

Thermal management also plays a critical role in maintaining performance stability [12,111,112]. High power density in compact transceiver modules can cause temperature-dependent changes in dielectric constant and loss tangent, leading to frequency detuning. Although thermally conductive fillers and ceramic substrates can mitigate heat buildup, they often introduce higher permittivity or increased processing complexity, creating performance trade-offs [55,113].

4.2. Engineering Challenges

From an engineering standpoint, cost and scalability are major barriers to widespread adoption. High-performance substrates typically rely on specialized materials and precision fabrication processes that increase production costs [68]. Efforts to develop low-cost hybrid laminates show promise, but maintaining consistent dielectric performance during mass production remains challenging due to process variability [56]. Integration with advanced packaging technologies, such as antenna-in-package architectures, further requires compatibility in thermal expansion and mechanical properties to ensure long-term reliability [56]. Long-term reliability under environmental stress, including humidity, thermal cycling, and mechanical strain, is another concern [83]. Material degradation can alter dielectric properties and affect system performance over time. Protective coatings and encapsulation techniques are commonly employed, but they may introduce additional loss or thickness constraints [53,83].

The mmWave transmission faces several more challenges than traditional low GHz radio frequencies due to its high propagation loss and limited link range. These difficulties arise from both propagation characteristics and technical constraints. The study in [50] identified key issues, including transceiver design complexity, high fabrication costs, reduced passive component performance at high frequencies, impedance matching requirements, and inadequate packaging. Although significant research efforts aim to reduce fabrication costs, additional problems remain [50]. The authors of [23] noted that mmWave substrate design incurs high mechanical manufacturing costs and faces efficiency losses due to the extreme frequency spectrum. Further challenges include manufacturing tolerances and process reliability, which affect system performance [23]. Therefore, these challenges underscore the necessity for integrated material innovation, advanced characterization, and scalable manufacturing strategies to fully realize the potential of substrates in next-generation mmWave communication systems. Table 6 provides a summary of these challenge, solution, and limitation analyses for mmWave substrates.

Table 6. Challenge, solution, and limitation analyses for mmWave substrates.

Category	Challenge	Current Approaches	Limitations	Impact on mmWave Systems
Technical	High dielectric and conductor loss	Low-loss polymers, ceramics, smoother copper	Cost, mechanical trade-offs	Reduced efficiency and link budget
Technical	Surface roughness effects	Rolled copper, roughness modeling	Adhesion concerns, process complexity	Increased insertion loss
Technical	Dielectric characterization uncertainty	Resonator and transmission line methods	Lack of standardization	Design uncertainty
Technical	Thermal instability	High thermal conductivity fillers	Increased permittivity	Frequency drift
Engineering	High material cost	Hybrid laminates	Performance compromise	Limited commercialization
Engineering	Manufacturing scalability	Automated lamination	Property variation	Yield issues
Engineering	Packaging integration	LTCC, AiP platforms	CTE mismatch	Reliability risks
Engineering	Environmental durability	Encapsulation, coatings	Added loss/thickness	Long-term degradation

5. Fabrication Techniques

Flexible electronics are rapidly advancing due to innovations in functional materials and fabrication methods, enabling the creation of uniquely shaped, cost-effective devices [49]. The emergence of 5G has further driven the development of RF devices operating at mmWave frequencies for use in wireless communication, automotive radar, high-resolution imaging, and gigabit networks [8]. These applications demand compact, high-density, low-loss, and cost-efficient system integration [8]. Therefore, substrate selection must consider factors like electrical loss, machineability, yield, and mechanical stability [8]. For instance, the authors of [114] illustrated the gains of deep learning-aided substrate shaping in mmWave communication systems utilizing dielectric resonator antenna configurations.

5.1. Various mmWave Fabrication Techniques

The section presents a review of various fabrication techniques, such as mmWave coating and chemical deposition, sputtering, and a variety of patterning and shaping techniques, such as photolithography, 3D printing, fabrication over mold, and new integrated techniques, including hot pressing for design or modification of substrate structures.

5.1.1. Composite Techniques

- Coating technique

Methods of coating, such as spin-coating, involve depositing particle-containing liquid mixtures onto a flexible substrate with apparatus such as brash or machine work and can be subsequently exposed to heat or allowed to dry to form uniform composite thin films [115]. For example, the authors of [116] have successfully produced a FeSiAl/WS2 soft magnet composite, which is a uniform coating of the micrometer-scale WS2 sheets on the FeSiAl grains; it is a typical low eddy current loss and medium effective magnetic permeability composite and is of high value for high-frequency applications.

- Chemical deposition technique

Chemical deposition includes the deposition of metals or oxides on flexible substrates by chemical solution from a chemical liquid under vacuum for the formation of a thin conductive film [115]. For instance, the authors of [117] prepared a magnetic polydimethylsiloxane (PDMS) hydrogel with chemical precipitation; the PDMS hydrogel was immersed in Fe^{2+}/Fe^{3+} solution, and then it was processed to uniformly disperse magnetic nanoparticles in the film.

- Sputtering technique

Sputtering is a thin-film deposition technique where high-energy particles bombard a target material in a vacuum, ejecting atoms that are deposited onto a substrate [115]. It enables the easy deposition of metals, alloys, and compounds [118]. For example, the authors of [119] used magnetron sputtering to deposit a 5 nm tantalum layer onto prestretched PDMS. Upon release, mismatched Young's modulus causes surface wrinkling, which was then used to sputter a 150 nm FeCoTa film, creating a flexible magnetic layer.

5.1.2. Patterned and Structured Techniques

- Photolithography

Photolithography is a widely used micro- and nanofabrication technique where patterns are transferred onto a substrate using photoresist and stepper tools, followed by etching to form microstructures, enabling precise structures at nanometer to micrometer scales, and includes variants such as soft, nanoimprint, e-beam, and scanning probe lithography [115]. For instance, the authors of [120] fabricated flexible 3D nanostructures like nanocoaxes on negative photoresist film through standard lithography stripping methods. The formed structures were chemically and mechanically stable and hence appealing for wearable and bioelectronics applications. Similarly, ref. [121] fabricated AgNW micropatterns based on PEG photolithography without universal etching, transferring on the hydrogels. The produced AgNW patterns exhibited good conductivity, flexibility, and water stability, which show possibilities for use in next-gen bioelectronics devices and implantable devices.

- Three-dimensional printing techniques

Three-dimensional printing technologies have been developed for manufacturing flexible electronic devices using conductive and resistive materials printed directly on flexible substrates, including polymers, papers, and silk [4,115]. Of these, FDM is one of the most popular, using extrusion of heated thermoplastic polymers, with resolution determined by the size of the nozzle [114]. For example, the authors of [122] applied FDM to fabricate flexible sensors, including a composite of polyvinylidene fluoride and cobalt ferrite, showing satisfactory both electrical and mechanical properties for wearable purposes. Direct writing methods can be employed for the creation of flexible metal circuits with high conductivity and flexibility as well [115]. The MXene inks, as utilized by the authors of [123], printed pressure sensors for touchpads and water level monitoring on flexible substrates that can also act responsively and accurately.

Yet another approach is Selective Laser Sintering (SLS), which builds structures by the layer-wise sintering of powder materials [116]. The authors of [124] fabricated a thermoplastic polyether block amide (TPE) and multi-walled carbon nanotube (MWCNTs)/nanocomposite sensor for sensitive and reliable E-skin health monitoring. Moreover, ref. [125] obtained the direct writing of the interfacial reaction of Polyvinyl Alcohol (PVA) and $AgNO_3$ as polymer-metal precursors for flexible circuits. Taken together, these methods indicate the growing potential of 3D printing and direct writing for fabricating customized, high-performance, and economically friendly flexible substrates.

- Fabrication with molds

Molding processes offer a simple and convenient approach for the interfacing of multiple materials, e.g., polymers, silicone, or other elastomers in a reusable mold and subsequently cured in the desired shape and profile [115]. They can shape a flexible structure, such as a substrate and soft robot unit, which has a defined shape, elastic–mechanical properties, and surface smoothness [115]. Shaped structures of custom configuration may be produced by changing the mold design. For instance, ref. [126] fabricated an earthworm-inspired slippage of microstructure-molding diaphragm by earthworm displacement, and was guaranteed to have a significant 3D bi-directional deformation in the low magnetic field of 40 mT attributed to its segmentation and radially magnetized validated structure. These could easily be adapted for integration with soft robotic systems. Similarly, ref. [127] fabricated a fast-response soft actuator by using Ecoflex rubber mixed with NdFeB particles. The actuator can be fabricated into grippers with different shape sizes, which exhibit some features of ultrafast response time (<0.1 s), ultra-large strain (<120%), long-term stability, and high precision, and is applicable for the grasping of objects of spherical and cylindrical shapes.

5.2. Integrated Techniques

Hot Pressing

Hot pressing (HP) is an inexpensive and easy process that can bond flexible base materials with electronic and conductive elements by means of controlled temperature and pressure with high precision, flexible electronic device [115]. Especially, it is a promising tool for modifying polymers and composites for improved physical properties [128]. The work in [128] also showed that HP is an effective approach to enhance electrical and thermal conductivity of acid-treated polymer PEDOT:PSS films to a high level of 1500 S/cm, 1.11 W/mK, respectively, owing to the optimized microstructure, thus making the HP-treated films be used as both a flexible conductor and heat sink. The authors of [129] developed a biodegradable, paper-like flexible composite using cellulose and graphene nanosheets. After hot pressing, the nanosheets were embedded into the fiber network, resulting in high isotropic conductivity (~10 Ω sq^{-1}), surpassing traditional conductive paper technologies. Therefore, hot pressing enhances material performance and supports the scalable, eco-friendly production of highly conductive and thermally efficient flexible electronics.

5.3. Other Fabrication Techniques

Other notable fabrication techniques for mmWave devices include the use of photoetchable glass (PEG) for creating substrate-integrated waveguide (SIW), as demonstrated by the authors of [130]. PEG's photoetchable properties enabled the formation of through glass via (TGVs), making it suitable as a dielectric material for SIW-based 5G applications. Similarly, glass substrates with TGVs have been adopted for packaging substrates and interposers in high-frequency applications [22,131], supporting components like filters, antennas, and waveguides [132]. Additionally, ref. [133] discussed the use of high-resistivity Fully Depleted Silicon-On-Insulator (FD-SOI) substrate for mmWave applications, focusing on material and fabrication challenges. Table 7 provides a summary of fabrication techniques for the creation of flexible devices.

Table 7. Overview of fabrication techniques: benefits, limitations, and cost implications.

Technique	Advantages	Disadvantages	Cost *
Coating	Simple and flexible to various materials	Limited thickness control	Low
Sputtering	Precise control of thickness and composition	High vacuum needed and slower deposition rate	Medium to High

Table 7. *Cont.*

Technique	Advantages	Disadvantages	Cost *
Photolithography	High resolution and precise designing	Not suitable for flexible substrates	High
3D Printing	Fast prototyping and customizable geometries	Material restrictions and surface roughness issues	Medium
Fabrication with Molds	Repeatable shapes and suitable for flexible substrates	Limited accuracy, wear and tear of molds over time	Low to Medium
Hot Pressing	Strong bonding and appropriate to multilayer structures	Needs heat-sensitive materials and irregular pressure may affect uniformity	Medium
SIW	Compact and easy integration with PCBs	Limited bandwidth and design difficulty	Medium
TGV	Excellent RF performance	Costly process and difficult orientation	High
FD-SOI	Low power consumption	Less mature ecosystem than bulk CMOS	Very High

* Cost Index (CI) on a 1–5 scale was used to compare fabrication technique costs based on weighted factors of material price, fabrication complexity, and availability, offering a transparent and consistent method to classify costs as low, medium to high, high, or very high.

6. Thermal Management

High-frequency operation in mmWave systems introduces not only elevated power dissipation but also localized thermal gradients arising from conductor losses, dielectric losses, and surface wave confinement [53,127]. These effects can lead to thermal failure mechanisms such as coefficient of thermal expansion (CTE) mismatch between the substrate, copper traces, solder joints, and packaging layers, resulting in mechanical stress accumulation, micro-cracking, delamination, or warpage during thermal cycling [21]. At frequencies above 24 GHz, even modest temperature rises can alter dielectric constant and loss tangent, thereby affecting impedance stability and radiation performance [134]. Furthermore, repeated thermal cycling can accelerate material fatigue and degrade adhesion between functional layers, particularly in multilayer laminates and flexible substrates [135].

From an engineering perspective, practical systems such as 5G base station antenna arrays often employ thermally conductive substrates (e.g., ceramic-filled laminates or aluminum nitride carriers) combined with heat spreaders and thermal interface materials to dissipate heat efficiently [136,137]. In one typical deployment scenario, high-power beamforming arrays integrate copper heat sinks and forced convection cooling to maintain temperature uniformity across antenna elements, preventing phase drift and performance degradation [138]. Another example is phased array front-end modules where, via stitching, metal backplanes, and thermal vias are used to reduce thermal resistance and minimize hot spots under high transmit duty cycles [139]. These approaches demonstrate how careful thermal design complements material selection to enhance reliability and ensure stable operation under continuous high-frequency loading.

6.1. Heat Dissipation Issues in mmWave Substrates

Material properties also affect the thermal management of mmWave substrates, for which the majority of the solutions are passive, with the substrate materials being closely tuned. Low thermal dissipation can be detrimental to the performance and reliability of electronic systems. The temperature can result in electrical property changes, which cause signal integrity problems, timing delays, and failure. Excessive operating temperature shortens the life of electronic components, degrades solder joints, and increases the potential of catastrophic failures. Decent heat dissipation performance is a crucial prerequisite to maintain normal operation level, low failure, and operation time of electronic devices [140]. Some researchers paid more attention to the important issue of heat dissipation

in mmWave substrates and devices and proposed advanced cooling and thermal management methods. The authors of [141] suggested a solution using a Schottky diode-integrated CVD diamond substrate to achieve improved heat dissipation and device power efficiency.

Another work [52] demonstrated that hybrid substrates enhance the thermo-mechanical reliability of mmWave modules. Furthermore, in [126], the authors show that AlN is much better than GaAs in enhancing the thermal characteristics of THz QCLs. Likewise, in [142], the authors idealized 3% Mo as the AlN:Mo ceramics refined for a δ-function heat under W-band radiation. The work in [143] presented a transparent radiative cooler based on 4H-SiC and stacked layers for passive cooling. Similarly, the work in [140] had c-BN films for the improvement of LED cooling, and the authors could decrease the temperature by 5–17 °C and increase the output by 27.9%. In conclusion, these reports, as summarized in Table 8, highlight the importance of high thermal conductivity materials, hybrid substrates, and passive cooling methods in better heat dissipation of advanced electronics.

Table 8. Summarized features of involved studies.

Reference	Substrate Materials	Thermal Conductivity	Temperature Range	Temperature Dependence	Cooling Approach
[52]	FR-4&hybrid	Not mentioned	−55 °C to +150 °C	Not mentioned	Passive (material selection)
[141]	CVD diamond	Not mentioned	Not mentioned	Not mentioned	Passive (substrate selection)
[142]	AIN:Mo	Not mentioned	Up to 1000 °C	Yes	Passive (material selection)
[143]	SiC with SiO_2	Not mentioned	54.3 °C to 29.1 °C	Yes	Passive (material selection)
[140]	c-BN&Si	c-BN > Si	Reduce from 5–17 °C	Yes	Passive (substrate selection)
[144]	AIN&GaAs	AIN > GaAs	93–95 K (op.), 80 K (heat)	Not mentioned	Passive (substrate selection)

6.2. Strategies for Enhancing Thermal Performance

Several studies have investigated successful methods to improve thermal performance of 5G mmWave substrates [111]. These techniques, as summarized in Table 9, are designed to answer the critical heat dissipation issues and to enhance mechanical robustness and system reliability. By taking these methods into account and applying suitable design principles, metal substrate PCBs can be engineered to provide efficient heat dissipation while ensuring reliability across a range of operating conditions.

Table 9. Methods for improving thermal performance.

S/N	Heat Dissipation Issues	Methods	Procedures
1.	Thermal stress	Component spacing	Providing adequate spacing between components helps prevent heat concentration and reduces the risk of thermal stress
		Thermal relief structures	Implementing thermal relief structures in copper planes or pads reduces the thermal expansion mismatch between components and the metal substrate, mitigating thermal stress
2.	Solder mask and surface finish	Solder mask material	Selecting a solder mask material with good thermal resistance properties helps withstand high operating temperatures and prevents delamination or cracking
		Surface finish	Opting for a surface finish that provides reliable solder joint formation and excellent thermal conductivity, such as ENIG (Electroless Nickel Immersion Gold), promotes robust thermal performance

Table 9. *Cont.*

S/N	Heat Dissipation Issues	Methods	Procedures
3.	Warping and distortion	Metal core thickness	Choosing an appropriate metal core thickness that balances mechanical stability and heat dissipation can help minimize warping issues
		Thermal expansion management	Understanding the CTE of the metal substrate and other PCB materials and ensuring compatibility can prevents warping and improves thermal stability
4.	Mechanical support and mounting techniques	Mounting holes	Incorporating sufficient mounting holes and using appropriate hardware, such as screws or standoffs, provides mechanical stability and prevents excessive stress on the PCB
		Shock and vibration resistance	Designing the PCB layout and selecting mounting techniques that offer resistance to shock and vibration ensures reliable operation in harsh environments

The thermal management in this study is spoken through a coupled electromagnetic thermal co-design framework, where power dissipation arising from conductor loss, dielectric loss, and radio frequency integrated circuit (RFIC) operation generates volumetric heat sources governed by the heat diffusion equation [145,146]:

$$\nabla \cdot (k \nabla T) + q = 0 \tag{1}$$

where ∇ (Del) is the vector differential operator that represents spatial derivatives, k is thermal conductivity, T is the temperature distribution in a material, and q is the volumetric heat generation rate. At mmWave frequencies, increased attenuation and current crowding elevate local heat density, while temperature-dependent permittivity and conductivity introduce feedback effects that can shift impedance matching and radiation characteristics through thermally induced dispersion and coupling variations [147,148]. In densely integrated architectures such as phased arrays and antenna-in-package (AiP) modules, thermal gradients and coefficient-of-thermal-expansion mismatch between substrates, metallization, and chips may lead to mechanical stress and performance drift, underscoring the need for substrates with optimized thermal conductivity, low dielectric loss, and stable electromagnetic properties for future high-frequency front-ends [149–151].

7. Electrical Performance

7.1. Effects of Substrate Materials on Signal Transmission Loss

The signal transmission speed V at high frequencies is influenced by transmission loss and interlayer dielectric resin, according to the authors of [19]:

$$V = \frac{kc}{\sqrt{\varepsilon_r}} \tag{2}$$

where c is the speed of light in a vacuum, k is a constant, and ε_r is the substrate's dielectric constant; lower ε_r results in faster signal propagation. At mmWave frequencies, even small variations in dielectric constant significantly influence phase velocity, dispersion, and timing accuracy, which directly affect signal integrity and system performance. Furthermore, transmission loss at mmWave frequencies becomes more pronounced due to several coupled mechanisms. Likewise, dielectric loss increases because molecular polarization cannot follow rapidly varying electric fields, leading to energy dissipation characterized by the loss tangent.

Additionally, conductor loss is amplified by the skin effect, which confines current to a very thin surface layer whose thickness decreases with frequency, thereby increasing effective resistance. Surface roughness of copper traces also contributes to additional scattering losses that are particularly critical above 20–30 GHz. Moreover, impedance mismatches and radiation losses become more sensitive as wavelengths shorten, making substrate permittivity stability essential. This section emphasizes that low-permittivity and low-loss substrates reduce electromagnetic energy storage and attenuation, enabling higher signal transmission speed, lower phase delay, and improved power efficiency in high-frequency interconnects and antenna structures. This is especially important for 5G and beyond systems operating in the mmWave spectrum, where material electromagnetic properties directly govern bandwidth capability and link reliability.

Additionally, in high-frequency circuits, the dielectric loss tangent (tan δ) of the substrate also affects signal loss, according to the authors of [19]:

$$\alpha = k' f \sqrt{\varepsilon_r} \tan \delta \tag{3}$$

where α is the transmission loss, k' denotes the phase constant, i.e., the real part of the complex propagation constant, which represents the wave number governing phase propagation along the transmission medium and is directly related to the effective permittivity of the substrate, and f is frequency; lower tan δ leads to reduced signal attenuation. Given signal speed and efficiency are closely related to the dielectric properties of copper-clad laminates [19].

Transmission of high frequency signal in PCBs is affected by problems such as the RC delay, crosstalk, energy loss, and impedance mismatch, which makes it difficult for PCBs to realize the miniaturization and signal integrity of performance [19]. These problems can be reduced by molecular structure design and the introduction of functional fillers, especially carbon nanofillers, which can easily control their dielectric properties according to morphologies, concentrations, and fabrication methods [19].

Various techniques have been proposed earlier for improving signal transmission at high frequencies. For example, the authors of [152] noticed that the dissipation factor is the one that affects the insertion loss to a great extent, and the dielectric constant has a smaller effect. The authors of [153] highlighted that reducing substrate loss is more effective with thinner dielectrics at lower frequencies. The silver-seed Semi Additive Process, as demonstrated by the authors of [154], yields smoother transmission lines that minimize conductor and overall transmission losses. Moreover, the authors of [155] showed that foam substrates provide the lowest signal loss and highest radiation efficiency and gain for antennas. Furthermore, ref. [156] confirmed that silver-seed copper plating on low-loss rigid substrates enhances conductor smoothness, further reducing signal loss. Together, these findings support the development of efficient, high-performance substrates for high-frequency electronic systems.

7.2. Mechanism of Dielectric Energy Loss

Dielectric loss occurs when molecular polarization cannot keep up with the oscillating electric field. If the relaxation time (τ) matches or is shorter than the field's frequency, the loss is minimal [19]. However, when the electric field oscillates faster than the polarization can respond, energy is absorbed and dissipated as heat [19]. Interestingly, when polarization becomes entirely unresponsive to the field, loss again becomes minimal [19]. The complex permittivity (ε^*) describes this behavior, comprising a real part (ε') representing dielectric constant and an imaginary part (ε'') representing dielectric loss as [19]:

$$\varepsilon^* = \varepsilon' - j\varepsilon'' \tag{4}$$

Their ratio yields the dielectric loss tangent as follows [19]:

$$\tan \alpha = \frac{\varepsilon''}{\varepsilon'} \tag{5}$$

Multiple studies have examined dielectric loss mechanisms in 5G mmWave systems, focusing on material characterization and loss reduction approaches. For example, authors of [78] emphasized that using low surface roughness copper foil significantly reduces conductor loss in transmission lines. In [37], the authors conducted broadband dielectric characterization up to 115 GHz, identifying low-dispersion, low-loss materials suitable for 5G. In [79], authors introduced new dielectric materials with low dissipation factors and strong adhesion to smooth copper foils, achieving notable reductions in transmission loss. Furthermore, ref. [83] developed a low-loss dielectric specifically for 5G and 6G mmWave applications. Taken together, the results of these studies provide guidance for dielectric loss mechanism optimization in high-frequency devices used in 5G wireless systems.

In detail, at mmWave frequencies, current is confined to a very thin surface layer of the conductor, making electromagnetic fields highly sensitive to metal surface topology. Increased surface roughness effectively lengthens the current path and enhances local field perturbations at the conductor–dielectric interface, which leads to higher conductor loss and increased effective dielectric loss due to stronger field interaction within the lossy substrate [157]. Therefore, reducing metal surface roughness improves signal propagation by minimizing conductor scattering and reducing excess attenuation, which is consistent with the theoretical framework described in [19]. The impact of conductor surface roughness on high-frequency attenuation has been widely documented and can be modeled using approaches mentioned in [157,158].

7.3. Signal Transmission Loss Reduction Procedures

Signal loss is one of the crucial issues in high-speed substrate, as it negatively affects system performance and reliability [112]. The conductor loss, dielectric loss, skin effect, radiation loss, impedance mismatch, and crosstalk are the main contributors. In order to solve these kinds of problems, the selection of low-loss dielectric materials and the refinement of copper foil types, such as the reverse-treated or the very-low-profile (VLP) foils, play a crucial role in conductor loss and signal integrity reduction at high frequency [159]. Several studies have proposed new mechanisms to combat the signal propagation losses in mmWave systems.

In [160], the authors proposed a novel contactless air-filled substrate integrated waveguide (CAF-SIW) bandpass filter in which the dielectric loss is suppressed by not using any dielectric materials, and the multilayer integration was realized. The work in [161] has recently proposed the thin-film integrated passive device (TFIPD) filters based on spiral inductors and metal–insulator–metal (MIM) capacitors, as well as a low insertion loss and high return loss for 5G bands. The authors of [162] coated PCBs with $SiCO_x/SiO_x$, and not only signal loss but also surface adhesion and thickness were improved.

Additionally, ref. [163] prepared epoxy-derived substrates with surface-modified aluminum flakes to enhance ε_r and reduce tan δ from 1 to 10 GHz. The research in [51] produced flexible copper-clad laminate (FCCLs) on polyimide based on direct metallization realized very low insertion loss (0.03–0.06 dB/mm up to 40 GHz), which outperforms adhesive counterparts. Likewise, the authors of [164] implemented a terahertz graphene antenna with a folded quarter-wave transformer and obtained a high gain (8.65 dB) and almost perfect radiation efficiency (99.8%).

These works show that reduced loss at mmWave frequencies calls for an approach that integrates low-loss materials, careful structural design, and advanced fabrication. Together, they emphasize the fundamental importance of material innovation, surface engineering, and architectural advancements for achieving the high performance required by future wireless sys-

tems such as 5G and terahertz systems. Table 10 summarizes salient points from the referenced studies on methods of reducing signal propagation losses in mmWave substrates.

Table 10. Summary of techniques to minimize signal transmission losses in mmWave substrates.

Reference	Substrate Type	Technique Used	Frequency Range	Achieved Reduction Loss
[51]	FCCL on polyimide	Direct metallization	0.1–40 GHz	Insertion loss of (0.03, 0.04, and 0.06) dB/mm at (20, 28, and 40) GHz, respectively
[160]	CLAF-SIW	Cavity-stacked filter design	mmWave	Enhanced filter performance with reduced insertion loss
[161]	GaAs-based TFIPD	Bandpass filters design	3.3–5.0 GHz	Insertion loss reduced to 0.755–1.237 dB
[162]	Multilayer $SiCO_x/SiO_x$ coating	Corrosion-resistant coatings	Not specified	Improved signal integrity
[163]	Epoxy with aluminum flakes	Stearic acid and Al_2O_3 coated aluminum flakes	1–10 GHz	Dielectric loss reduced from 0.037 to 0.005
[165]	Graphene	Circularly polarized 1 × 2 array antenna	2.345–2.555 THz	Achieved 99.8% radiation efficiency

7.4. Impact of Conductor Roughness in 5G mmWave Substrates

Recent studies highlight that the rapid expansion of mmWave applications for 5G systems introduces significant design challenges, particularly due to conductor surface roughness, which degrades signal transmission and antenna performance at high frequencies. Research by the authors of [164] demonstrated that increased surface roughness reduces conductivity and elevates loss as the operating frequency approaches and exceeds 30 GHz. Low-roughness copper foils, such as VLP, have proven effective in minimizing signal loss in thin mmWave substrates. Furthermore, models by the authors of [166,167] show a strong, often exponential, dependence of conductor loss on roughness, emphasizing the importance of advanced fabrication techniques to ensure reliable, low-loss transmission lines for future high-frequency 5G systems. Similarly, ref. [168] addresses the effects of surface roughness on wave propagation in the microwave and mmWave bands; this paper studies electromagnetic wave propagation and focuses on the propagation loss within an inner environment featuring surface roughness of the metallic waveguide structures, which can be applied using various commercial field solvers to handle surface roughness effects accurately and efficiently in any work involving conducting microwave structures.

7.4.1. Reduction in Transmission Line Conductor Loss

At mmWave frequencies (≥30 GHz), conductor surface roughness becomes a dominant contributor to insertion loss due to increased scattering and non-uniform current distribution at the conductor–dielectric interface [78,164]. When the root mean square roughness (RMS) roughness height (Δ) approaches or exceeds the skin depth (δ), typically 0.35–0.7 μm at 30–60 GHz for copper, the effective conductor resistance rises substantially [78]. The Hammerstad and Bekkadal roughness correction model yields the following [158]:

$$R_{rough} = R_{smooth}\left[1 + \frac{2}{\pi}\tan^{-1}\left(1.4\frac{\Delta}{\delta}\right)\right] \quad (6)$$

where R_{rough} (R_r) is the rough conductor surface resistance, $\tan^{-1}$ is the arctangent function (result in radians), Δ is the root mean square (RMS) surface roughness height, δ is the skin depth,

which is frequency-dependent, and $R_{smooth}(R_s)$ is the smooth-conductor surface resistance (per square) [78,168] given by the following:

$$R_s = \sqrt{\frac{\pi f \mu_o}{\sigma}} \tag{7}$$

where f is the frequency, μ_o the vacuum permeability, and σ copper conductivity (5.8×107 S/m). For copper, this yields (calculated values) δ (30 GHz) $\approx$ 0.382 µm, δ (40 GHz) $\approx$ 0.330 µm, and δ (60 GHz) $\approx$ 0.270 µm, computed from the following:

$$\delta = \sqrt{\frac{2}{\omega \mu_o \sigma}} \tag{8}$$

where ω is the angular frequency. To account for roughness, the Hammerstad and Bekkadal correction scales is used for the smooth R_s by a roughness factor F_r as follows [158]:

$$R_{s,rough} = F_r \times R_s, F_r = 1 + \frac{2}{\pi} \tan^{-1}\left(1.4\frac{\Delta}{\delta}\right) \tag{9}$$

The conductor loss in transmission lines or antenna feedlines is approximately proportional to $R_{s,rough}$, so percentage change in F_r maps directly to percentage change in conductor loss [158]. Using representative RMS roughness values obtained from Equation (6) for common copper foils, RA (rolled annealed): $\Delta \approx 1.8$ µm; LP (low profile): $\Delta \approx 0.9$ µm; and VLP (very low profile): $\Delta \approx 0.4$ µm. Applying the Hammerstad and Bekkadal formula (Equation (9)) produces the following factors and percent reductions (RA $\rightarrow$ VLP), as shown in Table 11. The percentage reduction is obtained by the following:

$$\% \text{ reduction } = \frac{(F_{r,RA} - F_{r,VLP})}{F_{r,RA}} \times 100\% \tag{10}$$

Table 11. Representative RMS roughness and estimated loss impact.

Frequency (GHz)	Skin Depth δ (µm)	(F_r) (RA, Δ = 1.8 µm)	(F_r) (VLP, Δ = 0.4 µm)	% Reduction in Conductor Loss (RA $\rightarrow$ VLP)
30	0.382	1.904	1.619	15.0%
40	0.330	1.917	1.661	13.4%
60	0.270	1.932	1.714	11.3%

The results in Table 11 show that, at mmWave, decreasing RMS roughness from RA $\rightarrow$ VLP reduces the roughness factor and thus conductor loss by roughly 11–15% across 30–60 GHz for the representative Δ values above. The absolute percent depends on Δ/δ: when $\Delta \gg \delta$, the benefit of further roughness reduction is reduced ($\tan^{-1}$ approaches $\pi/2$), so improvements saturate; when Δ is comparable to δ, improving surface smoothness is most effective. The practical implications for 5G and mmWave antenna and PCB design are as follows:

1. Choice of copper foil matters. Using VLP copper yields measurable reductions in conductor loss and, therefore, lower insertion loss in feeds and higher realized antenna gain/efficiency than RA foil.
2. Frequency dependence. Percent gain from smoother foil decreases as frequency rises beyond a point (because Δ/δ becomes very large and the correction saturates), but smoother foil is still beneficial in most 5G/mmWave bands.
3. Geometry and substrate coupling. Conductor loss is only one part of total loss; dielectric loss (tan δ), radiation efficiency, and conductor geometry (microstrip vs. CPW) also

matter; however, at mmWave, conductor roughness is often comparable in importance to dielectric loss for standard low-loss laminates.

4. Manufacturing and cost tradeoff. VLP foils are more expensive and can complicate processing (adhesion, etch characteristics). Designers must weigh 10–15% conductor-loss improvements against cost and process compatibility.
5. Mitigation strategies: Use VLP or electro-deposited ultra-smooth copper, adopt smoother plating/planarization, and use thicker conductors (reduces current crowding), optimize trace geometry, or use waveguide/SIW structures where conductor roughness is less critical.
6. This model predicts that moving from rolled-annealed (RA) copper (~1.8 μm RMS) to very-low-profile (VLP) copper (~0.4 μm RMS) can reduce conductor loss by about 25–35% at 30–40 GHz and up to 45% at 60 GHz, depending on substrate type and line geometry.

7.4.2. Computed Effect of Conductor Roughness on mmWave Transmission Loss

To evaluate the effect of conductor roughness, a 50 Ω microstrip line on a Rogers RO4350B substrate (ε_r = 3.48, tan δ = 0.0037, thickness = 0.254 mm) was examined over the 30–60 GHz range. The line width (W_{eff} = 0.56 mm) yields a standard 50 Ω impedance (Z_o). Using Equation (6), the total conductor loss (α_c) per unit length was estimated from [78,168]. Table 12 provides computed conductor loss cm^{-1}, and Figure 13 shows conductor loss against frequency for different copper foil roughness.

$$\alpha_c = \frac{8.686 R_{s,rough}}{Z_o W_{eff}} \tag{11}$$

where α_c is the total conductor loss, $R_{s,rough}$ is the rough conductor surface resistance, Z_o is the standard 50 Ω characteristic impedance, and W_{eff} is the effective line width of a conductor. The results clearly demonstrate that conductor surface roughness has a direct impact on signal loss at mmWave frequencies, where the skin depth becomes comparable to the texture of the copper surface. At 60 GHz, substituting RA copper (RMS ≈ 1.8 μm) with VLP copper (RMS ≈ 0.4 μm) reduces conductor loss from 0.094 dB/cm to 0.062 dB/cm, yielding approximately a 34% improvement.

Table 12. Computed conductor loss.

Frequency (GHz)	RA Copper (1.8 μm)	LP Copper (0.9 μm)	VLP Copper (0.4 μm)	Loss Reduction (RA → VLP)
30	0.071 dB/cm	0.060 dB/cm	0.051 dB/cm	28%
40	0.081 dB/cm	0.068 dB/cm	0.056 dB/cm	31%
60	0.094 dB/cm	0.078 dB/cm	0.062 dB/cm	34%

This decrease in loss contributes to lower insertion loss, higher radiation efficiency, and improved realized gain in antenna systems and feed networks. The influence of roughness increases with frequency as the ratio of roughness height to skin depth grows, intensifying electromagnetic scattering along the conductor–dielectric boundary. While ultra-smooth copper foils (RMS < 0.4 μm) or post-processing planarization can further reduce loss, these approaches incur additional fabrication cost. Therefore, using VLP or other low-roughness copper foils provides a practical balance between performance enhancement and manufacturability for 5G mmWave interconnects, phased array modules, and MIMO antenna substrates.

As illustrated in Figure 13, conductor loss rises with frequency across all foil types; however, rougher foils such as RA consistently show the highest loss, followed by LP, while VLP exhibits the lowest. This trend becomes increasingly pronounced at higher frequencies due to reduced skin depth. Thus, minimizing conductor roughness is a key requirement for achieving low-loss, high-efficiency mmWave designs.

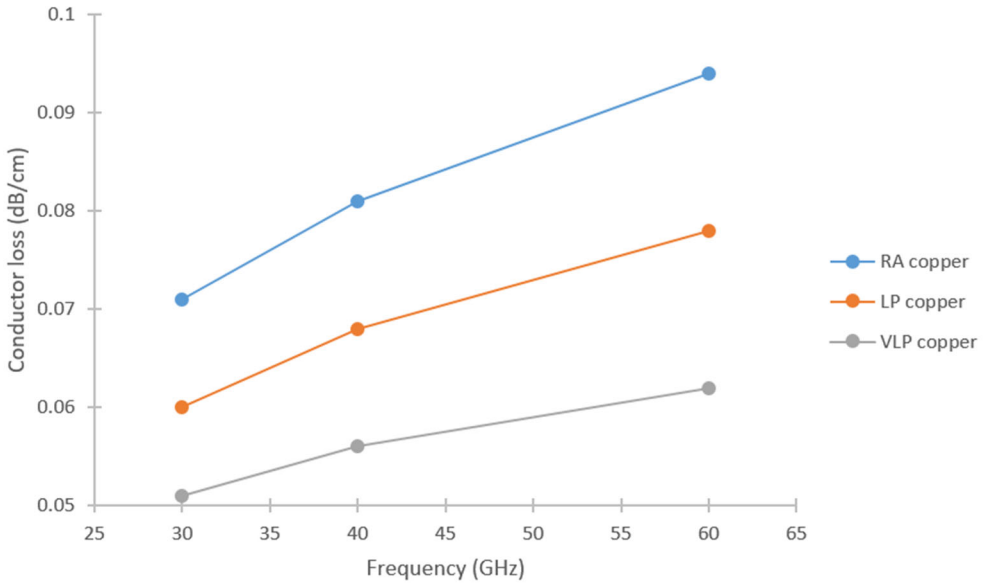


Figure 13. Conductor loss against frequency for different copper foil roughness.

7.4.3. Comparison of Theoretical Roughness Models and Experimental Observations at mmWave Frequencies

To strengthen the discussion on conductor surface roughness effects, a comparison between theoretical roughness models and representative experimental observations reported in recent mmWave studies [78,158,164,166–170] is given in this section. Measured transmission loss data for substrates employing rolled annealed (RA), low-profile (LP), and very-low-profile (VLP) copper foils across the 30–60 GHz range consistently demonstrate that increased surface roughness leads to higher insertion loss due to intensified current crowding, increased effective conductor path length, and enhanced scattering as the skin depth becomes comparable to roughness features [158,170–173].

While analytical formulations, such as the Hammerstad correction [158] and Huray snowball model [174], accurately predict the general trend [158,173–179], experimental results often exhibit slightly higher attenuation owing to practical factors including fabrication tolerances, copper grain morphology variations, surface oxidation, and measurement fixture uncertainties [158,169–173,180,181]. This synthesis of simulation and measurement evidence reinforces the importance of low-roughness conductor selection and precise material characterization in achieving low-loss performance for 5G and beyond-5G mmWave antenna substrates, thereby improving the robustness and practical relevance of this review study.

8. Mechanical Stability

Mechanical Properties of Substrate Materials

Scientific advancements in flexible electronics are increasing the demand for substrates that can endure repeated mechanical stress while maintaining reliability and performance. Numerous studies have explored strategies to enhance the mechanical properties of substrate materials for advanced mmWave applications. For instance, ref. [11] emphasized the need for substrate stability under frequent bending cycles, while [182] reviewed several innovative mechanical enhancement techniques. The authors of [183] demonstrated that a substrate's mechanical integrity directly affects strain and multilayer system reliability.

In other studies, such as [184], higher compliance and increased sensitivity and flexibility of the substrate by ferrite nanograins. The authors of [57] studied the evolution of grain structure at Al-Cu substrates of different thicknesses in EB-DED. Similarly, in [185], the MD method is employed to study the impact of growth conditions on SiNx film stresses and toughness.

The authors of [186] described an indentation approach for an accurate determination of the modulus of thin films on soft substrates. The work in [187] developed gradient nanocomposite coatings via halloysite nanotubes for well-tuned surface hardness and surface transitions. The research in [188] enhanced the hardness of ZnO thin film with Ag-Mn doping and annealing. The authors of [189] introduced chitosan-tetraethoxysilane (TEOS) hybrid coatings for adhesion and resistance improvement. Moreover, in [190], the authors proved that it is possible to suffer adhesion failure based on stress concentration when there is a modulus misfit between TiN and the substrate in the deposition, whereas ref. [191] optimized the ratios of Ag/Al to increase the hardness and reduce the friction in the Cu-Ag-Al coatings.

Collectively, these works illustrate the role of compositional design, surface engineering, and advanced fabrication approaches, including doping, gradient coating deposition strategy, simulation-guided growth, and hybrid layering for enabling tailoring of the substrate mechanical properties to the high-tech application. Table 13 summarizes the performance characteristics and heat dissipation concerns of various studies focusing on mmWave substrates.

Table 13. Comparative analysis of mechanical properties of mmWave substrate materials.

Reference	Material/Substrate	Mechanical Characteristics	Methods Applied	Key Findings
[57]	Al–Si–Cu/Al–Cu hybrid structures	Microstructure evolution with thickness; improved grain structure	Electron beam DED; hardness tests	Substrate thickness critical to mechanical robustness
[183]	PI and PET with Ag Nanowire Electrodes	High flexibility; elastic modulus PI ~5.27 GPa, PET ~3.31 GPa	Bending cycles up to 400,000	Minimal resistance increase; excellent strain accommodation
[184]	Ferrite-based nanoparticles on flexible substrate	Enhanced structural/electrical integrity under stress	Structural, optical, electrical characterization	Ferrite inclusion potentially supports improved mechanical endurance
[185]	SiN_x on SiO_2	Simulated high mechanical stability and adhesion	Molecular dynamics simulations	Potential use in high-performance, mechanically stable environments
[186]	2D materials on soft substrates	Measured Young's modulus, fracture strain, stress behavior	AFM, tensile and nanoindentation tests	Reveals mechanical limits of 2D-on-soft system
[187]	Gradient nanocomposite on soft substrate	Enhanced surface hardness and adhesion	Nanoindentation, SEM	Gradient coatings improve mechanical resilience
[188]	Ag–Mn doped ZnO on glass	Structural change due to heating; mechanical property alteration	XRD, SEM, nanoindentation	Thermal treatment modulates mechanical properties
[189]	Hybrid chitosan-silica on Al	Improved toughness and flexibility	Mechanical and chemical testing	Biocomposite coatings enhance durability
[190]	Multilayer TiN	Failure mechanisms under mechanical stress analyzed	Micro/nano-mechanical analysis	Interfacial mechanics critical for durability
[191]	Cu-alloy on Al	Hardness and strength dependent on microstructure	Laser deposition and microhardness testing	Laser parameters critically influence mechanical behavior

9. Hybrid mmWave Substrates Materials

In recent years, the evolution of information transmission technologies has driven electronic devices toward miniaturization, high frequency, and faster data rates [19]. The thin-film technologies have emerged as vital tools in this transition, offering high-resolution patterning suitable for mmWave applications and enabling integration of high-performance passive components [192]. Hybrid substrate materials have drawn much attention in the literature, inspired by their low dielectric loss, high thermal stabilities and the potential for 5G mmWave systems. For example, ref. [56] prepared thermosetting siloxane hybrid (Singi GTM-1 and 2) doped with quartz prepreg, and the composite exhibited low transmission loss at 10 GHz. The authors of [193] fabricated a cyclic olefin copolymer (COC) based substrate with an ultra-low dissipation factor for mmWave and THz applications. Likewise, in [184], the authors created a flexible metamaterial sensor by means of $Ca_xCo_{(0.90-x)}Zn_{0.10}Fe_2O_4$ in FR-4 for chemical sensing.

In other related work, as in [55], the authors realized the transmission lines and filter with Aluminum Ribbon Ceramic (ARC), showing a low insertion loss and high performance. Furthermore, in [194], the authors used aerosol jet-printed alumina for leaky wave antennas, which has very good resilience. In [195], the authors demonstrated MEMS switches on quartz and silicon with <3 dB insertion loss covering up to 750 GHz. Additionally, in [196], the authors used the Graphene-Assembled Film (GAF) to develop a 128 antenna array with good bandwidth and low sidelobes. Additionally, in [197], the authors demonstrated the bilayer $LiNbO_3$ on silicon/sapphire for high-Q acoustic resonators, and a study by the authors of [198] proposed a sub-6 GHz 5G high-gain rectangular patch antenna with multilayer superstrates.

All of these analyses bring attention to the fact that hybrid-formed substrates like COC, ARC, GAF, siloxane-quartz, and $LiNbO_3$ stacks are imperative to achieve low loss, high performances and multi-functionality for forthcoming mmWave 5G systems. These materials demonstrated excellent performance in high-frequency applications, including antennas, filters, sensors, switches, and resonators, supporting high data throughput, low attenuation, compactness, and mechanical reliability. Table 14 provides a comparative summary of these hybrid substrates.

Table 14. Summary of hybrid materials for mmWave substrates.

Reference	Substrate Materials	Frequency Band	Possible Applications
[55]	Alumina ribbon ceramic	24–100 GHz	5G and mmWave communication
[56]	Thermosetting siloxane hybrid	Not explicitly mentioned	Advanced electronic packaging
[63]	Alumina ribbon ceramic	mmWave	High-performance filters for 5G small cells
[184]	Ferrite-based nanoparticles	Not specified	Chemical sensing
[193]	Low-loss material (PTFE or similar)	mmWave and THz	mmWave/THz communication systems
[194]	Thin alumina	mmWave	Microstrip leaky wave antenna
[195]	Silicon-based for MEMS	THz	MEMS switch design
[196]	Graphene-based	5G (Wideband)	For 5G mobile communication
[197]	Lithium niobate ($LiNbO_3$)	50.74 GHz	For mmWave front-end filters
[198]	Rogers RT5880 and RT5870	5.2 GHz	For sub-6 GHz 5G applications

Design Principles and Performance Enhancement Mechanisms of Hybrid mmWave Substrates

Hybrid mmWave substrate materials have emerged as a strategic approach to reconcile the competing requirements of low dielectric loss, mechanical flexibility, thermal stability, and scalable fabrication. The innovation mechanism of hybrid substrates is rooted in the synergistic

combination of organic polymer matrices with inorganic fillers or frameworks, enabling simultaneous tuning of electromagnetic and thermo-mechanical properties. At frequencies above 24 GHz, where dielectric dispersion, surface wave effects, and dimensional tolerances strongly influence antenna efficiency, hybridization allows precise engineering of effective permittivity and loss tangent while improving structural reliability [189,193,195].

From an electromagnetic perspective, organic matrices such as PTFE, LCP, or epoxy offer inherently low dielectric constants and reduced dipolar relaxation losses, which are advantageous for minimizing signal attenuation [18,163,199]. However, these materials typically suffer from low thermal conductivity and limited dimensional stability. Incorporating inorganic constituents such as quartz, silica, ceramic nanoparticles, or glass fibers introduces low-loss dielectric pathways and suppresses molecular motion, thereby reducing dielectric dispersion and improving phase stability at mmWave frequencies [55,106,189].

From an electromagnetic standpoint, the effective medium behavior can be tailored through filler volume fraction, particle morphology, and interfacial bonding, allowing precise control of ε_r and tan δ; hence, the effective dielectric constant ε_{eff} of hybrid composites can be estimated using mixing rules derived from effective medium theory. A commonly used approximation is the Maxwell–Garnett formulation [200,201]:

$$\varepsilon_{eff} = \varepsilon_m \left(\frac{\varepsilon_f + 2\varepsilon_m + 2f\left(\varepsilon_f - \varepsilon_m\right)}{\varepsilon_f + 2\varepsilon_m - f\left(\varepsilon_f - \varepsilon_m\right)} \right) \tag{12}$$

where ε_m is the permittivity of the matrix, ε_f is the filler permittivity, and f is the filler volume fraction.

This relation illustrates how dielectric properties can be tuned through compositional control to achieve target impedance and phase velocity characteristics for mmWave antennas [201,202]. For higher filler loadings or interconnected microstructures, the Bruggeman symmetric model [203] provides a more accurate description:

$$f\frac{\varepsilon_f - \varepsilon_{eff}}{\varepsilon_f + 2\varepsilon_{eff}} + (1-f)\frac{\varepsilon_m - \varepsilon_{eff}}{\varepsilon_m + 2\varepsilon_{eff}} = 0 \tag{13}$$

These models highlight the role of microstructural topology in determining dielectric response and guiding material optimization for low-loss transmission lines and radiating elements [202,203].

Mechanical reinforcement is another key mechanism. Organic–inorganic interfaces distribute mechanical stress and reduce thermal expansion mismatch, which is particularly important for antenna substrates subjected to temperature cycling and high RF power densities [52,187]. Hybrid networks also improve resistance to warpage and microcracking, maintaining consistent impedance characteristics across wide bandwidths [162]. Furthermore, engineered interfaces reduce moisture absorption pathways, which otherwise lead to dielectric degradation at high frequencies [75].

Organic matrices such as PTFE, LCP, and modified epoxies provide low intrinsic dielectric loss due to reduced dipolar relaxation [18,199], while inorganic fillers like silica, quartz, and ceramic nanoparticles suppress molecular mobility and reduce dielectric dispersion [61,106]. The resulting composite exhibits improved phase stability and reduced attenuation across mmWave bands. Interfacial engineering through surface functionalization or coupling agents minimizes interfacial polarization losses, which can otherwise become significant at tens of gigahertz [64,204].

Mechanical and thermal enhancements arise from load transfer across the organic-inorganic interface, reducing the coefficient of thermal expansion mismatch between conductive layers and substrates. This is critical for maintaining impedance stability in phased arrays exposed

to thermal cycling and RF heating [135,189]. Hybrid architectures also improve dimensional stability and resistance to warpage, preserving radiation pattern integrity and minimizing detuning effects [57,189].

Emerging hybrid systems such as graphene-reinforced composites introduce high thermal conductivity pathways via phonon transport while maintaining low dielectric loss when operated below electrical percolation thresholds [205,206]. These materials improve heat dissipation in high-power antenna modules and reduce thermal gradients that could otherwise induce performance drift [1,113]. Similarly, siloxane–quartz hybrids combine molecular flexibility with ultra-low dielectric loss, offering excellent environmental stability and reduced moisture uptake [56].

Microstructural optimization, including nanoscale dispersion, graded filler distributions, and multilayer hybrid laminates, further enhances performance by reducing scattering losses and enabling tailored electromagnetic field confinement. Such strategies support the development of compact, high-efficiency antenna substrates suitable for dense integration in 5G and future mmWave systems [52]. Generally, hybrid substrates embody a materials-by-design paradigm in which electromagnetic performance, thermal management, and mechanical robustness are co-optimized through controlled composition and architecture. Their continued advancement is expected to underpin reliable high-frequency antenna platforms and scalable manufacturing pathways for next-generation wireless infrastructure.

Furthermore, the hybrid substrate platforms can be interpreted through effective medium theory as mentioned above, where the composite permittivity and loss behavior arise from electromagnetic field interactions across heterogeneous phases, influencing wave propagation constant, dispersion, and surface current distribution [200,201,207]. Organic–inorganic hybridization, including graphene-reinforced composites and siloxane–quartz systems, enables simultaneous reduction in dielectric loss tangent and enhancement of thermal pathways, thereby mitigating loss-induced heating and suppressing parasitic coupling in high-density interconnect environments [52,187]. Such materials support improved field confinement and reduced radiation leakage in AiP and phased array implementations, while providing mechanical compliance that alleviates stress under thermal cycling, making them promising candidates for scalable mmWave and emerging terahertz modules [187,208].

10. AI-Driven RSSI-Aware Beamforming and Implications for 5G mmWave Substrate Design

Recent advances in 5G and emerging 6G mmWave systems have demonstrated the increasing importance of artificial intelligence (AI) techniques, particularly received signal strength indicator (RSSI)-aware beamforming and deep learning-based beam management, in addressing the challenges of highly directional propagation, blockage sensitivity, and rapid channel variation at frequencies above 24 GHz [6,26,27,209–212]. While AI-driven beamforming improves link adaptation, its effectiveness is inherently dependent on the electromagnetic stability and loss characteristics of antenna substrates, highlighting the need for integrated algorithm material co-design. Survey work by the authors of [210] highlights the importance of auxiliary information (e.g., positioning and signal measurements) in reducing beam training overhead, emphasizing that RSSI-assisted strategies can significantly accelerate beam alignment. Security-oriented research, such as [26] BeamSecure-AI, further expands the scope by demonstrating how AI can monitor beam-level anomalies using signal features, including received power patterns, to enhance resilience in mmWave radio access networks.

Deep learning approaches have been widely investigated for joint beam and power optimization. A study by the authors of [211] proposes a learning-based framework that integrates channel features to improve beam and power allocation efficiency. Strengthening learning methods, such as those by the authors of [213], enables adaptive beam management and

relay selection in vehicular mmWave communications, addressing fast channel dynamics. Significantly, ref. [27] clearly incorporate RSSI measurements into deep learning models for hybrid beamforming, demonstrating that signal strength feedback can guide beam selection with reduced training overhead.

Emerging applications extend AI beamforming to specialized environments. Work by the authors of [214] applies deep learning to RIS-assisted maritime networks, highlighting robustness in challenging propagation conditions, while [215] discovers combined learning for secure beamforming in future THz systems, indicating the scalability of AI-based approaches. Moreover, ref. [216] provides a broader perspective on machine learning-enhanced smart antenna systems, reinforcing the role of data-driven optimization.

Complementary hardware-oriented studies emphasize the physical layer context in which AI beamforming operates. The study by the authors of [6] establishes how substrate properties influence phased-array performance, ref. [212] analyzes mechanical effects such as wrinkling on electromagnetic behavior, and ref. [217] presents practical beam-steerable antenna arrays. These works underline that accurate beamforming decisions, including RSSI interpretation, depend on underlying antenna and material characteristics.

The reviewed studies in this section jointly show that RSSI-aware AI techniques improve beam selection accuracy, reduce training latency, enhance robustness to mobility, and enable adaptive resource management in mmWave networks. However, they also reveal ongoing challenges, including generalization across environments, integration with hardware constraints, and the need for standardized evaluation frameworks. AI beamforming performance is fundamentally constrained by RF front-end losses, dielectric stability, and conductor roughness; therefore, including it in this study provides cross-layer context linking intelligent algorithms with substrate engineering, as shown in Figure 14 as well as in Table 15.

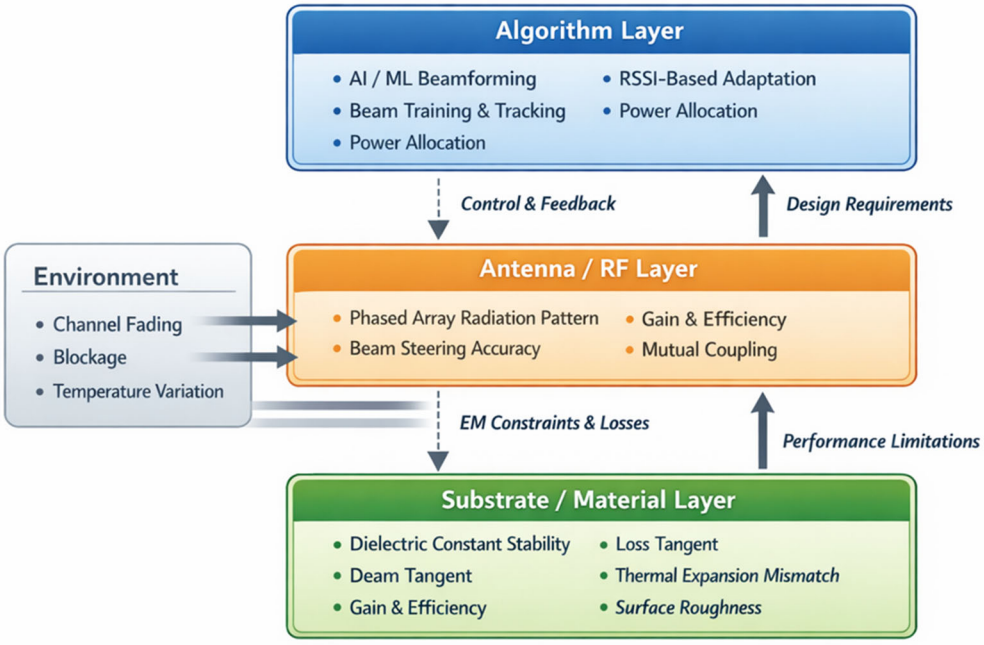


Figure 14. Cross-layer co-design framework linking AI beamforming, antenna, and substrate.

Table 15. Gap and contribution mapping for RSSI-aware/AI beamforming literature.

Ref.	Key Contribution	Relevance to 5G mmWave Substrate
[6]	Experimental substrate performance study	Directly supports 5G mmWave substrate design
[26]	AI-based beam anomaly detection	Found how AI can monitor beam-level anomalies using signal features to enhance resilience in mmWave radio access networks

Table 15. *Cont.*

Ref.	Key Contribution	Relevance to 5G mmWave Substrate
[27]	RSSI-driven deep learning beam selection	Directly tied to RF signal integrity affected by substrate properties
[209]	AI-aided 3D beamforming using spatial learning	Displays need for stable antenna and substrate performance for accurate beam control
[210]	Survey of positioning/RSSI-assisted beam training	Connects propagation sensitivity to hardware/material design
[211]	Deep learning optimization framework	Shows performance depends on RF front-end losses
[212]	EM prediction under structural changes	Provides substrate reliability
[213]	DRL for beam management	Highlights need for low-loss substrates for reliable links
[214]	DL beam selection for RIS systems	Demonstrates propagation sensitivity
[215]	Federated learning beamforming	Discovers combined learning for secure beamforming in future THz systems
[216]	ML-enhanced smart antennas	Links algorithms to antenna hardware
[217]	Active antenna array demonstration	Deals with hardware context

11. Conclusions

In 5G and mmWave antenna design, substrate selection plays a central role in determining key performance characteristics such as gain, bandwidth, radiation efficiency, impedance stability, and overall device reliability. This work underscores the importance of material engineering, advanced electromagnetic design, and accurate system-level modeling in developing substrates capable of supporting high-frequency operation in emerging 5G and beyond communication systems. The mmWave spectrum offers the advantage of large bandwidth and extremely high data rates; however, achieving efficient radiation at frequencies above 24 GHz requires substrates with low dielectric constants to reduce surface wave propagation and low loss tangents to limit dielectric energy dissipation.

Among the conventional high-frequency laminates, Rogers RT/Duroid 5880, Arlon DiClad 880, and Rogers CLTE-MW are extensively utilized due to their low permittivity, low loss tangent, and robust thermal and environmental stability. For instance, Rogers RT/Duroid 5880 ($\varepsilon_r \approx 2.20$, $\tan\delta \approx 0.0009$) demonstrates excellent radiation efficiency beyond 77 GHz with minimal moisture absorption and a stable thermal coefficient of permittivity. Arlon DiClad 880 ($\varepsilon_r \approx 2.17$, $\tan\delta \approx 0.0009$) features smooth copper surfaces that reduce conductor losses and maintain stable impedance up to 90 GHz. Rogers CLTE-MW ($\varepsilon_r \approx 2.94$, $\tan\delta \approx 0.0015$) exhibits strong dimensional stability, making it suitable for demanding mmWave environments such as 60 GHz WLAN and short-range radar systems.

Alongside these traditional substrates, next-generation materials are gaining attention. Graphene-based substrates offer reconfigurability, high conductivity, and tunable dielectric response, making them promising for adaptive and beam-steerable antennas. Flexible substrates such as Liquid Crystal Polymer (LCP) support compact and wearable 5G devices due to their mechanical flexibility, low loss ($\tan\delta \approx 0.002$), and minimal moisture absorption. Meanwhile, Low-Temperature Co-Fired Ceramics (LTCC), with dielectric constants of 5.8–7.0, provide multilayer circuit integration and thermal stability, making them suitable for compact phased arrays and automotive radar around 77 GHz.

Overall, Rogers RT/Duroid 5880, Arlon DiClad 880, and Rogers CLTE-MW remain highly suitable for mmWave systems due to their dielectric performance and reliability, while graphene, LCP, and LTCC materials represent the future pathway toward flexible, tunable, and highly integrated 5G and beyond communication technologies.

12. Future Outlook and Research Gaps

This section combines the key research gaps identified throughout the manuscript and outlines priority directions for future research. Ongoing research into novel substrate materials, improved fabrication techniques, and enhanced modeling is vital to overcome existing limitations. Standardization and environmental stability considerations will also be critical for widespread deployment. The continuous advancement of substrate technologies for 5G and mmWave systems is driving efforts toward high-performance, miniaturized, and cost-efficient antenna and circuit designs.

Despite significant progress in low-loss materials, fabrication technologies, and hybrid substrate concepts, several unanswered challenges remain, including the lack of standardized dielectric characterization methods above 60 GHz, limited understanding of long-term thermal–mechanical reliability under high-frequency operation, and scalability constraints in manufacturing advanced hybrid substrates. Moreover, conductor surface roughness effects, environmental stability, and integration with emerging AI-driven adaptive systems require further study. Addressing these gaps will be necessary for facilitating robust, improved surface finishing, modeling techniques, high-efficiency mmWave antenna platforms, and supporting next-generation wireless systems. Future work should also explore more on AI-assisted substrate optimization, eco-friendly materials, development of low-loss, thermally stable, and cost-effective materials such as graphene composites, LCPs, and CLTE-MW to enhance antenna efficiency, integration, and reliability testing under variable environmental conditions to achieve stable, scalable, and high-performance substrates for emerging 5G and 6G applications.

The summarized research gaps, as shown in Table 16, derived from the critical analyses presented across this study particularly the discussions on technical and engineering challenges (Section 4), thermal mechanisms and reliability considerations (Section 6), conductor roughness modeling and experimental limitations (Section 7), and hybrid substrate innovation pathways (Section 9) highlight key directions for advancing mmWave substrate technologies. Unlike prior surveys that primarily address antenna structures or material developments separately, this study provides a comprehensive and physics-grounded synthesis integrating dielectric behavior, conductor effects, thermal reliability, fabrication constraints, and emerging hybrid material strategies within a unified context. Addressing these gaps will be essential for enabling reliable ultra-high-frequency platforms and supporting the evolution toward 6G systems, where extreme bandwidths, sub-THz operation, and intelligent adaptive networks will require tightly coordinated advances across materials, devices, and system architectures.

Table 16. Summary of research gaps.

Area	Identified Gap	Technical Significance	Scope for Investigation
Dielectric Characterization	Lack of standardized measurement above 60 GHz	Inaccurate material models affect antenna prediction	Develop high-frequency metrology standards
Material Loss Mechanisms	Limited understanding of frequency-dependent losses	Impacts link budget and efficiency	Multi-physics modeling and broadband measurements
Thermal Reliability	Insufficient data on long-term thermal cycling effects	Causes drift and failure in phased arrays	Accelerated aging studies and thermal design optimization
Conductor Roughness	Model–measurement discrepancies at mmWave	Leads to uncertainty in loss estimation	Experimental validation and improved roughness models
Hybrid Substrates	Trade-off between flexibility and low loss not fully optimized	Limits practical deployment	Advanced composite design and nano-fillers
Fabrication Precision	Tolerance control at sub-millimeter scales	Affects beam accuracy	High-precision manufacturing techniques
Environmental Stability	Moisture and temperature sensitivity	Performance variation in field conditions	Protective coatings and robust material systems

Table 16. *Cont.*

Area	Identified Gap	Technical Significance	Scope for Investigation
Scalability	High cost of advanced materials	Barrier to commercialization	Cost-effective processing methods
System Integration	Limited co-design between materials and beamforming algorithms	Suboptimal performance	Cross-layer optimization frameworks
Reliability Modeling	Lack of predictive lifetime models	Difficult maintenance planning	Physics-of-failure modeling

Hence, from a forward-looking perspective in this study, future 6G and THz platforms will require substrate solutions that simultaneously manage electromagnetic dispersion, thermal transport, and multi-physics coupling across increasingly integrated architectures. At sub-THz frequencies, wavelength shrinkage amplifies sensitivity to surface roughness, dielectric inhomogeneity, and interconnect parasitics, while higher power densities intensify heat generation and thermo-mechanical interactions that can degrade link stability and beamforming accuracy [218–220]. Progress in cross-layer co-optimization spanning antenna arrays, RF circuits, packaging, and materials engineering will therefore be critical to achieve ultra-low-loss propagation, stable beam steering, and reliable operation in next-generation wireless systems.

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